A project report on

SELF HEALING NEURAL NETWORKS

Submitted in partial fulfillment for the award of the degree of

Bachelor of Technology
Computer Science & Engineering

by

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VIT-AP UNIVERSITY

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DECLARATION

I hereby declare that the thesis entitled “SELF HEALING NEURAL NETWORKS ” submitted by me, for the award of the degree of Bachelor of Technology in Computer Science and Engineering by VIT is a record of bonafide work carried out by me under the supervision of Dr. Shaik Kareemulla.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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This is to certify that the Senior Design Project titled "**SELF HEALING NEURAL NETWORKS**" that is being submitted by **VEDANT KANOJE (22BCE9039)** is in partial fulfillment of the requirements for the award of Bachelor of Technology, is a record of bonafide work done under my guidance. The contents of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for award of any degree or diploma and the same is certified.

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ABSTRACT

Neural networks have achieved remarkable performance in image classification, natural language processing, autonomous systems, and decision-making applications. However, despite their success, modern neural networks remain highly vulnerable to internal faults, noisy perturbations, adversarial manipulations, and unexpected operational degradation. Traditional robustness methods such as adversarial training and static regularization techniques are limited because they cannot dynamically adapt once the model is deployed.

This project presents a Self-Healing Neural Network (SHNN) framework capable of autonomously detecting faults and recovering degraded model performance without requiring complete retraining. The proposed framework introduces a closed-loop architecture consisting of fault injection, anomaly detection, fault localization, and automated healing mechanisms.

A baseline neural network was trained on the MNIST dataset using TensorFlow and Keras. A modular fault injection engine was developed to simulate multiple realistic failure scenarios including weight corruption, neuron failure, layer degradation, and random parameter perturbations. A monitoring and detection module was implemented to track internal activations, confidence shifts, and output degradation during inference.

Upon fault detection, self-healing strategies such as weight restoration, localized fine-tuning, and adaptive parameter correction were applied to recover model performance. Experimental results demonstrate that the proposed framework successfully detects internal faults and restores a significant portion of the lost accuracy while minimizing computational overhead.

The developed framework contributes toward more resilient and autonomous AI systems suitable for safety-critical and long-running applications where reliability and adaptability are essential.

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INTRODUCTION

1.1 Background and Motivation

Artificial Intelligence and deep learning systems are now being used in many real-world applications like healthcare, autonomous vehicles, cybersecurity, finance, industrial automation, and smart communication systems. Deep Neural Networks (DNNs) have shown very strong performance in areas such as image recognition, speech processing, pattern analysis, and decision making because they can learn complex features from very large datasets. But even though these models are powerful, they are still quite sensitive to faults, noisy inputs, hardware instability, parameter corruption, and unexpected conditions during deployment.

Most traditional neural networks are designed with the assumption that once training is completed, the model will continue working properly during deployment. In real-world conditions this assumption often does not hold properly. Neural networks working in practical systems can face memory corruption, noisy sensor data, hardware degradation, malicious perturbations, and changes in input distributions. Even a small disturbance in network weights or neuron activations can reduce prediction accuracy and overall reliability by a large amount. Research has shown that even tiny internal faults or perturbations may cause serious prediction failures, specially in safety critical applications.

Existing robustness techniques such as adversarial training, regularization, retraining, and static defense methods improve reliability only for predefined fault conditions. Most of these methods depend heavily on prior knowledge of attacks or failures and are not very adaptive once the model is deployed. Also, retraining a neural network after deployment takes high computational resources and is not practical for many real-time systems.

Inspired from biological self-repair systems and self-healing mechanisms used in communication networks and integrated circuits, researchers introduced the idea of self-healing neural networks. These systems continuously monitor their internal behaviour, detect abnormal conditions, and automatically apply corrective actions during runtime. Instead of depending fully on retraining or

manual intervention, a self-healing framework attempts to recover stability through adaptive recovery techniques.

The main motivation behind this project is to build a practical self-healing framework capable of simulating neural network faults, detecting degradation in model behaviour, and applying automated recovery strategies. The proposed framework combines fault injection, monitoring, detection, and healing into a single closed-loop pipeline. A graphical user interface is also included for interactive experimentation and visualization of fault and healing behaviour.

1.2 Problem Statement

Even with the rapid progress in deep learning technologies, modern neural networks are still highly fragile when exposed to unexpected operational disturbances. Small corruption in weights, neuron failures, noisy parameter updates, or structural perturbations can reduce model accuracy and make predictions unstable. Most existing robustness approaches are static and mainly depend on assumptions made during training. Because of this, deployed neural networks usually cannot detect internal failures or recover automatically once degradation starts happening.

Current approaches like adversarial training and full model retraining also have several practical limitations. They require large computational resources, depend on prior knowledge of attack patterns, are difficult to perform continuously during runtime, and cannot adapt properly to unseen failures. This creates a major gap between theoretical robustness methods and the actual needs of real-world intelligent systems where continuous reliability and autonomous recovery are important.

The main problem addressed in this project is the lack of a lightweight and practical framework that can simulate realistic neural network faults, automatically detect degradation, identify vulnerable network components, apply runtime healing methods, and recover model performance without complete retraining. To solve this issue, the project proposes a self-healing neural network framework with integrated fault detection and automated recovery mechanisms working in a closed-loop manner.

1.3 Research Objectives

The primary objective of this project is to design and implement a self-healing neural network system capable of automatically detecting and recovering from internal faults and parameter degradation.

The major objectives of the project are listed below:

1. To develop a baseline neural network model for image classification using deep learning techniques.
2. To design a modular fault injection engine capable of introducing controlled faults such as weight noise, neuron failure, weight corruption, layer degradation, and random weight swapping.
3. To study the effect of different fault types, severity levels, and layer-wise corruption on neural network performance.
4. To implement monitoring and fault detection techniques using activation statistics, confidence analysis, threshold-based anomaly detection, and performance deviation tracking.
5. To develop self-healing strategies for recovering degraded neural networks through rollback-based restoration, localized repair, and adaptive recovery operations.
6. To integrate all modules into a unified closed-loop self-healing framework.
7. To build a graphical user interface for interactive experimentation, monitoring, and visualization of recovery behaviour.
8. To experimentally evaluate the robustness, recovery capability, and fault tolerance of the proposed framework under multiple fault conditions.

1.4 Scope of the Work

This project mainly focuses on developing a software-based self-healing framework for neural networks working under simulated fault conditions. The work targets classification models implemented using TensorFlow and Keras on the MNIST handwritten digit dataset.

The scope of the project includes development of a baseline neural network, implementation of controlled fault injection techniques, anomaly monitoring, automated healing methods, GUI-based visualization, and experimental evaluation under different fault conditions. The project focuses more on internal neural network degradation rather than only external cybersecurity attacks. Different fault severities and layer-specific corruption experiments are analysed to better understand the robustness behaviour of neural architectures.

The work does not focus on large-scale distributed neural systems, transformer-based language models, hardware-level autonomous repair, or production-scale deployment systems. Instead, the project aims to provide a foundational framework that can later be extended for future research in self-healing AI systems.

1.5 Contributions of the Project

The important contributions of this project are summarized below:

- Development of a modular fault injection framework supporting multiple neural fault types.
- Experimental analysis of layer-wise fault sensitivity in neural networks.
- Design and implementation of a closed-loop fault detection and self-healing pipeline.
- Development of automated rollback-based restoration techniques for runtime healing.
- Integration of monitoring, detection, healing, and visualization into a unified framework.
- Development of a graphical user interface for real-time experimentation and analysis.
- Detailed evaluation of fault severity, recovery behaviour, and layer vulnerability under multiple experimental conditions.

The project shows that neural network degradation can be detected and partially recovered through autonomous runtime intervention without requiring complete retraining.

1.6 Organization of the Report

This report is divided into six chapters.

Chapter 1 introduces the background, motivation, objectives, scope, and overall contributions of the project.

Chapter 2 discusses the literature review related to neural network robustness, self-healing systems, fault detection, and recovery techniques.

Chapter 3 explains the proposed self-healing framework including system architecture, fault injection engine, monitoring pipeline, and healing methods.

Chapter 4 describes the implementation details, algorithms used, development environment, dataset information, and GUI integration.

Chapter 5 presents the experimental results and analysis including layer-wise sensitivity analysis, severity-based degradation behaviour, and healing performance evaluation.

Chapter 6 concludes the report by summarizing the major findings, discussing limitations, and suggesting possible future improvements for self-healing neural network systems.

Literature Review

2.1 Neural Network Vulnerability and Failure Modes

Deep Neural Networks (DNNs) are being used in many fields such as image recognition, medical diagnosis, natural language processing, autonomous systems, and cybersecurity. These models have achieved very high accuracy in many tasks, but they are still sensitive to different kinds of faults and disturbances. Even small changes in weights, neuron activations, or input data can reduce prediction accuracy and overall model stability.

Researchers have identified many different types of neural network failures. Some common failure modes include adversarial attacks, noisy inputs, parameter corruption, hardware faults, gradient instability, and distributional changes in data. Among these, adversarial attacks are one of the most studied problems. In these attacks, very small perturbations are added to inputs in such a way that humans cannot easily notice them, but the neural network gives wrong predictions. Studies showed that even tiny modifications can create major classification errors in well-trained models.

Apart from external attacks, internal faults inside neural networks can also create serious degradation. Problems like neuron failure, corrupted weights, and structural disturbances can damage learned feature representations and affect decision boundaries. Some experimental studies found that dense layers and output layers are more sensitive to corruption and parameter noise compared to other layers.

Traditional neural network training methods mostly assume stable deployment conditions and do not include runtime self-repair capabilities. Because of this, modern neural networks still remain fragile when deployed in uncertain environments where conditions keep changing continuously.

2.2 Self-Healing Systems in Artificial Intelligence

The idea of self-healing systems originally came from biological immune systems and fault-tolerant engineering designs. A self-healing system can detect abnormal behaviour, diagnose faults, and automatically apply corrective actions to restore normal operation.

Recently, researchers started applying self-healing concepts in artificial intelligence systems and communication networks. Self-healing AI mainly focuses on building systems that can adapt continuously, recover during runtime, and maintain stability without requiring human intervention.

Chen et al. proposed a closed-loop framework for self-healing neural networks where the system continuously monitors internal model behaviour and applies corrective signals during inference. Their research showed that dynamic feedback-based recovery can improve robustness against unexpected perturbations better than some static defense methods.

Similarly, studies in self-healing communication networks showed that intelligent monitoring and adaptive recovery systems can automatically detect faults and restore service continuity using machine learning techniques. These studies highlighted the importance of closed-loop architectures where fault detection and repair work together continuously.

Overall, self-healing AI systems represent a shift from static machine learning models toward more adaptive and resilient intelligent systems capable of maintaining operational stability during runtime.

2.3 Fault Detection in Neural Networks

Fault detection is one of the most important parts of a self-healing neural network system. The main purpose of fault detection is to identify abnormal model behaviour before complete performance degradation happens.

Many approaches have been proposed for neural network fault detection. Some commonly used methods include statistical monitoring, anomaly detection, activation analysis, confidence monitoring, and reconstruction-based techniques.

Threshold-based statistical monitoring is one of the simplest and most commonly used approaches. In this method, runtime statistics like activation mean, variance, confidence scores, and gradient norms are compared with baseline healthy model behaviour. If the deviation becomes too high, the system considers it as a possible fault or degradation condition.

Researchers have also explored anomaly detection methods using autoencoders and reconstruction models. These techniques try to reconstruct healthy activation patterns and detect abnormalities using reconstruction error values. If the reconstruction error becomes large, it may indicate internal instability inside the neural network.

Some studies showed that internal feature representations can provide strong indicators of neural network instability and failure conditions. However, many existing detection methods are still computationally expensive and difficult to use in real-time systems.

Recent research suggests that effective fault detection systems should work continuously during inference while maintaining low computational overhead and high sensitivity to faults.

2.4 Neural Network Repair and Recovery Methods

Neural network repair mainly focuses on restoring model performance after degradation or failure occurs. Existing repair methods can generally be divided into four categories: retraining-based repair, parameter restoration, architectural repair, and adaptive runtime correction. Traditional retraining methods attempt to recover performance by training the network again using clean or augmented datasets. Although retraining is effective in many cases, it requires high computational cost and is not suitable for real-time systems.

Research on neural network repair identified several common repair techniques such as correcting layer dimensions, restoring connectivity, adjusting activation behaviour, modifying optimization functions, and reconstructing corrupted parameters.

DeepPatch introduced a maintenance-oriented approach where adaptive processing units were inserted into the neural network to selectively correct degraded feature representations during inference. This method improved runtime robustness without reducing baseline classification accuracy too much.

Other researchers proposed rollback-based recovery methods where corrupted parameters are partially restored using previously saved healthy checkpoints. Compared to full retraining, rollback-based recovery requires much lower computational cost while still allowing partial recovery from faults.

Localized fine-tuning methods were also explored in some studies. In these approaches, only the faulty layers are retrained while the remaining network stays frozen. This reduces recovery time and helps preserve previously learned feature representations.

Even with these advancements, many repair techniques still face several limitations such as high computational overhead, dependency on predefined fault conditions, limited runtime adaptability, and difficulty maintaining both robustness and standard accuracy together.

2.5 Limitations of Existing Approaches

Although significant progress has been made in neural network robustness and repair research, many existing approaches still have important limitations.

1. Static Defense Methods

Most robustness techniques are designed during training itself and cannot adapt dynamically after deployment. Methods like adversarial training and preprocessing mainly protect against already known fault conditions.

2. High Computational Cost

Complete retraining and adversarial fine-tuning require large computational resources, making them difficult to use in real-time or low-resource environments.

3. Lack of Runtime Recovery

Many systems can detect degradation but cannot automatically recover during runtime. Human intervention is still required in many repair processes.

4. Poor Adaptability to Unknown Faults

Several methods are optimized only for specific attack patterns or perturbation types and may not work properly under unseen operating conditions.

5. Accuracy and Robustness Tradeoff

Research often shows that improving robustness can reduce standard classification accuracy. Maintaining both robustness and baseline performance together is still a difficult challenge.

6. Limited Internal Layer Analysis

Many studies focus mainly on external adversarial attacks while giving less attention to internal neural failures such as neuron corruption, parameter instability, and layer degradation.

These limitations show the need for practical self-healing systems capable of runtime monitoring, adaptive recovery, lightweight operation, and autonomous fault handling.

2.6 Research Gap and Motivation for Proposed Framework

The literature review shows that neural networks are still highly vulnerable to both internal and external faults even after many advances in robustness research. Most existing methods focus mainly on static defense techniques and depend heavily on expensive retraining procedures. Very few systems provide integrated closed-loop architectures capable of performing fault injection, runtime monitoring, automated detection, and autonomous healing together.

In addition, many existing studies mainly focus on adversarial robustness and provide limited analysis of layer-wise sensitivity, severity-based degradation, sequential corruption behaviour, and runtime recovery mechanisms.

The reviewed research also shows a lack of lightweight experimental frameworks where users can interactively evaluate neural network degradation and self-healing behaviour under controlled fault conditions.

To address these limitations, this project proposes a modular self-healing neural network framework integrating controlled fault injection, statistical fault detection, rollback-based healing, and GUI-driven experimentation. The proposed system focuses on runtime adaptability

and autonomous recovery while experimentally analysing neural network behaviour under multiple fault scenarios and severity levels. The framework aims to provide a practical base for future research in resilient and self-adaptive AI systems.

Proposed Self-Healing Framework

3.1 System Overview

The proposed framework introduces a modular self-healing neural network system that can automatically detect and recover from internal model degradation caused by injected faults and parameter corruption. The framework is designed as a closed-loop recovery pipeline where fault simulation, monitoring, detection, and healing mechanisms work together continuously during execution.

The main goal of the framework is to improve neural network robustness against runtime disturbances without requiring complete retraining or manual intervention. The system continuously monitors internal model behaviour, identifies abnormal conditions, and applies recovery operations whenever performance degradation crosses predefined threshold values.

The framework mainly consists of five major components:

1. Baseline Neural Network Model

2. Fault Injection Engine

3. Fault Detection Module

4. Self-Healing and Recovery Module

5. Graphical User Interface (GUI)

These modules together form a complete experimental platform for analysing neural network robustness, fault sensitivity, and automated recovery behaviour.

The proposed architecture follows self-healing concepts discussed in previous neural repair and closed-loop AI research where detection and recovery happen dynamically during runtime instead of relying only on static protection methods.

3.2 Overall Architecture of the Framework

The complete framework architecture is organised as a sequential closed-loop pipeline.

Input Data

↓

Baseline Neural Network

↓

Fault Injection Engine

↓

Fault Detection Module

↓

Performance Evaluation

↓

Healing Trigger

↓

Recovery Mechanism

↓

Re-evaluation

↓

Recovered Model Output

The architecture works in several stages.

First, the baseline neural network processes clean input data under normal operating conditions. Then controlled faults are injected into selected layers using the fault injection engine. After fault injection, internal monitoring systems analyse runtime behaviour and compare it with stored healthy reference statistics. If deviations become larger than predefined thresholds, the fault detection module marks the network as degraded. The healing module then applies recovery operations such as rollback restoration or localized repair. Finally, the recovered model is evaluated again to check whether operational stability has been restored properly.

This closed-loop structure allows autonomous diagnosis and recovery without needing full model retraining every time faults occur.

3.3 Baseline Neural Network Model

A baseline neural network model was developed using TensorFlow and Keras frameworks. This model acts as the healthy reference system against which all degraded and recovered states are compared.

The framework uses the MNIST handwritten digit dataset because it is computationally simple, easy for fast experimentation, and suitable for controlled fault analysis.

The implemented neural network architecture contains:

one Flatten input layer, one Dense hidden layer with 128 neurons, one Dropout layer, and one Dense output layer with 10 neurons.

Baseline Model Structure

Flatten- Converts image matrix into vector

Dense 128- Feature extraction and learning

Dropout- Regularization

Dense 10- Final classification output

The model was trained using Adam optimizer, Sparse Categorical Cross-Entropy loss function, and accuracy as the main evaluation metric.

The trained baseline model achieved nearly 97.5% classification accuracy on the MNIST test dataset, which was used as the healthy operational benchmark during all experiments.

The baseline model also provided healthy activation statistics, weight distributions, and confidence behaviour which were later used for fault detection and healing evaluation.

3.4 Fault Injection Engine

The fault injection engine is one of the most important modules of the proposed framework. Its main purpose is to simulate realistic neural network degradation under controlled experimental conditions.

The engine was designed as a modular and extensible system capable of injecting different fault types into selected neural network layers with varying severity levels.

Design Objectives

The fault injection engine was designed to: support controlled experimentation, evaluate layer-wise sensitivity, simulate runtime degradation, and generate fault conditions for testing healing mechanisms.

The framework implements five major fault types.

3.4.1 Weight Noise Injection

Weight noise injection adds Gaussian random noise into network weights.

W_{faulty}=W+\mathcal{N}(0,\sigma)

Here, (W) represents original weights and (\sigma) represents fault severity. This fault simulates unstable parameter updates, analog hardware noise, and stochastic corruption. Experimental analysis showed that high noise severity causes rapid instability and major accuracy degradation.

3.4.2 Weight Zero Fault

This fault randomly sets a percentage of network weights to zero.

W_{faulty}=W\odot M

Here, (M) represents a binary mask matrix. This fault simulates sparse hardware failure, memory corruption, and parameter dropout. Experimental results showed that moderate sparsity could still be tolerated because neural networks have some level of redundancy.

3.4.3 Neuron Failure Injection

Neuron failure injection disables complete neurons by zeroing entire columns of Dense layer weight matrices. This fault simulates dead neurons, processing unit failure, and inactive computational pathways. Experimental observations showed that output-layer neuron failures caused severe classification collapse, especially at higher severity levels.

3.4.4 Layer Degradation Fault

Layer degradation scales layer weights downward proportionally.

W_{faulty}=(1-s)W

Here, (s) represents degradation severity. This fault simulates gradual hardware aging, signal attenuation, and precision loss. The model showed good resilience against moderate degradation until near-complete weight collapse occurred.

3.4.5 Random Weight Swap Fault

This fault randomly swaps weight positions inside layers. The operation damages learned structural organisation and destroys feature relationships learned during training. Experimental analysis showed that random swap faults caused near-total model failure even at lower severity levels.

3.5 Fault Detection Module

The fault detection module continuously monitors neural network behaviour during inference and identifies abnormal operating conditions. The module compares runtime behaviour with baseline healthy statistics collected during initial training.

The system monitors: activation distributions, prediction accuracy, loss variation, and parameter deviation.

Threshold-Based Detection

The framework uses threshold-based anomaly detection.

|x_{runtime}-x_{baseline}|>\tau

Here, (x_{runtime}) represents runtime statistics, (x_{baseline}) represents healthy reference behaviour, and (\tau) is the detection threshold. If runtime deviation becomes larger than the threshold value, the model is classified as degraded.

The detection module was designed to remain lightweight while still providing good sensitivity toward internal neural degradation.

3.6 Self-Healing and Recovery Mechanisms

Once a fault is detected, the healing module tries to restore network stability and recover classification accuracy. The framework mainly uses rollback-based recovery methods inspired by checkpoint restoration and neural repair research.

3.6.1 Rollback-Based Weight Restoration

The main recovery strategy partially restores corrupted weights using previously saved healthy checkpoints.

W_{recovered}=\alpha W_{healthy}+(1-\alpha)W_{faulty}

Here, (W_{healthy}) represents stored healthy weights, (W_{faulty}) represents corrupted weights, and (\alpha) controls rollback intensity. This method allows lightweight recovery, preserves already learned features, and restores performance faster compared to complete retraining.

3.6.2 Localized Layer Repair

Instead of retraining the complete network, only the detected faulty layers are repaired selectively. This helps reduce computational overhead, recovery latency, and unnecessary parameter changes. Layer-specific healing was chosen because experimental analysis showed that Dense layers are much more vulnerable compared to non-trainable layers.

3.7 Closed-Loop Recovery Pipeline

The proposed framework works as a continuous closed-loop recovery system.

Recovery Pipeline

Healthy Model

↓

Fault Injection

↓

Performance Degradation

↓

Detection Module

↓

Fault Localization

↓

Healing Trigger

↓

Recovery Execution

↓

Re-evaluation

↓

Recovered Operation

This feedback-driven architecture allows continuous monitoring, autonomous adaptation, and runtime recovery. The design was inspired by closed-loop fault-tolerant systems and self-healing integrated circuit architectures.

3.8 Graphical User Interface Design

A graphical user interface (GUI) was developed to simplify experimentation and visualization of the proposed framework. The GUI allows users to train models, inject faults interactively, select fault types, adjust severity levels, monitor degradation, and trigger healing operations easily.

GUI Features: Feature Description, Model Training, Fault Selection, Severity Control, Layer Selection, Choose target layers, Detection Visualization, Result Monitoring

3.9 Workflow of the Complete System

The complete workflow of the proposed framework is summarized below.

System Workflow

1. Train baseline neural network.

2. Save healthy model checkpoints and baseline statistics.

3. Inject selected faults into target layers.

4. Evaluate degraded model performance.

5. Monitor internal runtime statistics.

6. Detect abnormal behaviour using threshold analysis.

7. Trigger healing mechanism.

8. Restore corrupted parameters using rollback recovery.

9. Re-evaluate the healed model.

10. Display results and performance metrics through GUI.

The proposed framework provides a practical and extensible foundation for self-healing neural network research by integrating controlled fault simulation, runtime monitoring, autonomous detection, and adaptive recovery into one unified experimental platform.

IMPLEMENTATION DETAILS

4.1 Development Environment and Tools

The proposed self-healing neural network framework was implemented using Python and different deep learning libraries. The implementation mainly focused on modularity, easy experimentation, and reproducibility.

Software used: Python 3, TensorFlow, Keras, NumPy, Tkinter, Matplotlib, TensorFlow Datasets, Google Colab, VS Code

TensorFlow and Keras were selected because they provide simple neural network construction, flexible layer manipulation, and efficient training support. NumPy was mainly used for matrix operations, random fault generation, and parameter modification.

4.2 Dataset Preparation and Preprocessing

The framework uses the MNIST handwritten digit dataset for all experiments. The dataset contains 60,000 training images and 10,000 testing images of handwritten digits from 0–9. Each image has a size of 28×28 pixels.

The preprocessing pipeline included:

1. Loading the dataset using TensorFlow datasets.

2. Normalizing pixel values to the range [0,1].

3. Splitting training and testing datasets.

4. Feeding normalized inputs into the neural network.

Normalization was done using: x_{normalized}=\frac{x}{255.0}

Here, (x) represents original pixel values. Normalization improved convergence speed, gradient stability, and numerical consistency during training.

The dataset was intentionally kept simple so controlled analysis of degradation and recovery behaviour could be performed without very high computational complexity.

4.3 Baseline Model Training

The baseline neural network was implemented using the TensorFlow Keras Sequential API.

Model Architecture: Flatten - Input shape: (28×28)

Dense - 128 neurons, ReLU activation

Dropout - 0.2 dropout rate

Dense - 10 output neurons|

The hidden layer used the ReLU activation function. f(x)=\max(0,x)

ReLU was selected because it gives fast convergence and reduces vanishing gradient problems. The output layer generated logits which were later converted into probabilities using Softmax.

The loss function used was sparse categorical cross-entropy.

L=-\sum_{i=1}^{N} y_i\log(\hat{y}_i)

Here, (y_i) represents true labels and (\hat{y}_i) represents predicted probabilities. Adam optimizer was used because of its adaptive learning rate and fast convergence behaviour.

Training Configuration: Optimizer - Adam, Epochs - 5,| Loss Function- Sparse Categorical Crossentropy

After training, the model achieved nearly 97.5% classification accuracy on the MNIST testing dataset. The trained weights were saved and later used as healthy checkpoints for rollback recovery and fault comparison.

4.4 Implementation of Detection Mechanisms

The fault detection system continuously monitored neural network behaviour during inference. The module was designed to identify degradation early and trigger healing automatically.

The monitored indicators included: prediction accuracy, output confidence, loss increase, activation deviation, and parameter changes.

The framework compared runtime statistics with healthy baseline values. Detection was based on deviation analysis.

D=|x_{runtime}-x_{baseline}| if: D>\tau

the model was classified as degraded. The threshold value was adjusted experimentally to balance sensitivity and false positive rates.

4.5 Implementation of Healing Strategies

The framework mainly implemented rollback-based healing and localized recovery methods.

Rollback Recovery

Rollback recovery restored corrupted parameters using healthy checkpoints.

recovered = alpha * healthy + (1 - alpha) * faulty

Here, `alpha` controls restoration intensity. This approach preserved learned features while reducing computational overhead compared to full retraining.

Localized Recovery

Instead of retraining the entire network, only faulty layers were repaired while unaffected layers remained unchanged. This reduced recovery latency and unnecessary parameter modification.

The healing module improved degraded model performance significantly under moderate fault conditions.

4.6 GUI Integration and User Interaction

A graphical user interface was developed using Tkinter to provide interactive access to framework functions. The GUI was directly connected with training modules, fault injection systems, detection mechanisms, and healing operations.

The GUI improved usability and allowed experiments to be performed without changing source code repeatedly.

4.7 Experimental Configuration

The experimental setup was designed to evaluate layer-wise vulnerability, severity-based degradation, sequential corruption behaviour, and healing performance.

Experimental Categories

The framework tested:

1. Single-layer fault injection

2. Multi-layer sequential injection

3. Severity-based degradation

4. Layer sensitivity analysis

5. Healing effectiveness analysis

Severity Levels: 0.15-Low fault intensity, 0.40-Moderate degradation,0.70-Severe corruption, 1.00- Complete failure

Evaluation Metrics

The main evaluation metrics were: classification accuracy, loss increase, F1 Score, ROC-AUC Scores.

RESULTS AND ANALYSIS

The proposed self-healing neural network framework was evaluated using a multilayer perceptron (MLP) trained on the MNIST handwritten digit dataset. The experimental pipeline implemented a complete closed-loop architecture consisting of fault injection, anomaly detection, and automated healing modules. The system was designed to evaluate how different neural faults affect model reliability and how effectively recovery mechanisms can restore degraded performance.

The baseline neural network achieved a classification accuracy of approximately 97.47% with a baseline loss of 0.085 before fault injection. The architecture consisted of a Flatten input layer, a Dense-128 hidden layer with ReLU activation, a Dropout layer, and a Dense-10 output layer.

The developed framework implemented five fault types:

- Weight Noise
- Weight Zero
- Neuron Failure
- Layer Degradation
- Random Weight Swap

These faults were injected at varying severity levels and across different layers using the modular fault injection engine designed for the project.

The framework additionally implemented:

- statistical fault detection using activation deviation monitoring,
- threshold-based anomaly analysis,
- F1-score evaluation,
- ROC-AUC evaluation,
- rollback blend healing,
- layer repair healing.

The experimental results demonstrate that neural network robustness is strongly dependent on both fault type and layer location.

Baseline Performance

Before fault injection, the model achieved stable classification performance:

Metric	Value
Baseline Accuracy	97.47%
Baseline Loss	0.085
F1 Macro	0.9659
F1 Weighted	0.9660
ROC-AUC Macro	0.9990
ROC-AUC Weighted	0.9990

The baseline ROC curves showed near-perfect separability across all ten MNIST classes, confirming that the network had learned highly stable feature representations before degradation.

Layer-Wise Fault Sensitivity Analysis

Layer-wise experiments revealed substantial differences in fault tolerance across the architecture.

The Flatten layer and Dropout layer exhibited almost complete resilience to injected faults because they contain either no trainable parameters or inactive regularization during inference. In contrast, the Dense-128 hidden layer and Dense-10 output layer showed high vulnerability.

The Dense-10 output layer emerged as the most critical failure point in the architecture. Corruptions in this layer directly affected class logits and probability distributions, causing severe prediction instability.

The Dense-128 hidden layer showed high sensitivity to structural corruption faults because it is responsible for latent feature extraction and intermediate representation learning.

The overall layer sensitivity ranking observed experimentally was:

Rank	Layer	Sensitivity
1	Flatten	Minimal
2	Dropout	Minimal
3	Dense-128	High
4	Dense-10	Critical

The results confirm that computational dense layers are significantly more fault-sensitive than structural layers.

Fault Severity Analysis

Faults were injected at severity levels ranging from 0.15 to 1.00 to evaluate progressive degradation behavior.

Neuron Failure

Neuron failure produced approximately linear degradation as more neurons were disabled.

Severity	Accuracy
0.15	~88%
0.40	72.71%
0.70	38.07%
1.00	9.80%

Complete neuron shutdown caused total model collapse, demonstrating strong dependence on active neuron pathways.

Weight Noise

Weight noise produced the most unstable behavior among all tested faults.

Severity	Accuracy	Loss
0.40	44.96%	6.0951
0.70	19.79%	50.7314

1.00	13.38%	134.9908

The results showed exponential loss growth under noisy perturbations, indicating that stochastic weight corruption severely disrupts learned feature representations.

Weight Zero

Weight zeroing demonstrated that the network possesses partial redundancy.

Severity	Accuracy
0.40	92.91%
0.70	69.52%
1.00	9.80%

Moderate sparsity was tolerated relatively well before critical connectivity loss occurred.

Layer Degradation

Layer degradation produced minimal impact until complete degradation.

Severity	Accuracy

0.40	97.52%
0.70	97.54%
1.00	9.80%

The network demonstrated high tolerance to proportional attenuation of weights.

Random Swap

Random weight swapping was the most destructive structural fault.

Severity	**Accuracy**
0.15	~18%
0.40	9.36%
1.00	9.83%

Even low swap severity caused near-random predictions because learned weight organization was destroyed.

The observed destructive severity ranking was: Random Swap > Weight Noise > Neuron Failure > Weight Zero > Layer Degradation

Sequential Multi-Layer Injection Results

Sequential fault injection experiments were performed by progressively corrupting multiple layers simultaneously.

The results showed cumulative degradation behavior.

Weight Noise

Injection Stage	Accuracy
L0 + L1	48.89%
L0 + L1 + L2	48.60%
L0 + L1 + L2 + L3	18.06%

Neuron Failure

Injection Stage	Accuracy
L0 + L1	90.96%
L0 + L1 + L2	93.16%
L0 + L1 + L2 + L3	51.63%

Random Swap

Injection Stage	Accuracy
L0 + L1	64.86%
L0 + L1 + L2	10.39%
L0 + L1 + L2 + L3	9.78%

The experiments demonstrated that corruption of the final output layer consistently caused the largest accuracy collapse.

Fault Detection Performance

The framework implemented statistical activation monitoring for anomaly detection using configurable threshold and sample-size parameters.

The detection module compared baseline layer activations against current activations during inference. Layers exceeding the configured threshold were flagged as anomalous.

In experiments involving neuron failure across Dense layers:

- the detection module successfully identified corrupted layers,
- activation deviation values increased significantly in damaged layers,
- detection remained stable across multiple injected fault types.

The threshold parameter controlled anomaly sensitivity, while the sample-size parameter stabilized statistical estimates across inference batches. This enabled robust detection without requiring retraining of the detection system.

The experiments confirmed that activation-based monitoring can effectively identify internal neural degradation in real time.

- Dense layers were highly fault-sensitive.
- Output layers showed maximum degradation.
- Random weight swaps caused catastrophic failure.
- Layer degradation faults were comparatively less destructive.
- Rollback recovery partially restored model accuracy in moderate fault conditions.

These results validated the effectiveness of the proposed self-healing framework under controlled neural fault scenarios.

Healing and Recovery Results

The framework implemented two healing strategies:

1. Rollback Blend
2. Layer Repair

Rollback Blend restored damaged weights toward saved baseline weights using a configurable blend factor.

The healing process significantly restored model performance after severe degradation.

One representative experiment injected neuron failure into Dense layers, reducing model accuracy to approximately 63.3%. After rollback blend healing with blend strength 0.5, the healed model recovered to approximately 97.1% accuracy.

State	Accuracy
Baseline	97.47%
Faulty	63.3%

Healed	97.1%

The healed model additionally achieved:

Metric	Value
F1 Macro	0.9586
F1 Weighted	0.9589
ROC-AUC Macro	0.9987
ROC-AUC Weighted	0.9988

The recovery process restored both classification accuracy and class separability across all digit classes.

The rollback blend method proved highly effective because it preserved portions of the learned baseline representations while simultaneously reducing corrupted parameter influence.

F1 Score and ROC-AUC Evaluation

F1-score and ROC-AUC metrics were integrated into the framework to evaluate multiclass classification robustness under faults.

Under severe faults:

- F1 macro scores dropped below 0.1 in catastrophic corruption scenarios,

- ROC-AUC values dropped toward random classification levels (~0.45–0.50),
- class imbalance effects became more visible during output-layer corruption.

After healing:

- F1 scores recovered close to baseline,
- ROC curves regained near-perfect separability,
- class-wise prediction stability improved substantially.

These results confirm that the healing system restored not only overall accuracy but also balanced multiclass predictive behavior.

Comparative Analysis with Existing Research

The experimental observations align closely with prior research in self-healing networks and neural repair systems.

Previous work on AI-based self-healing systems emphasizes autonomous fault detection and recovery as critical requirements for resilient intelligent systems.

Research on DNN repair patterns identified network connectivity corruption and data-dimension failures as major reliability challenges in deep learning systems.

DeepPatch demonstrated that maintaining robustness while preserving baseline accuracy is a major challenge in neural repair systems.

The implemented rollback blend approach in this project similarly demonstrated that selective restoration mechanisms can recover robustness while minimizing accuracy loss.

The implemented statistical monitoring methodology additionally aligns with experimental sensitivity analysis techniques discussed in controlled experiment research.

FINAL DISCUSSION

The experimental results demonstrate that neural networks exhibit highly non-uniform fault sensitivity. Structural layers such as Flatten and Dropout showed near-complete resilience, while dense computational layers exhibited substantial vulnerability to corruption.

Among all tested faults:

- random structural corruption was the most destructive,
- stochastic weight noise caused severe instability,
- sparse parameter loss was comparatively tolerable,
- proportional degradation produced minimal impact until complete collapse.

The proposed framework successfully demonstrated:

- controlled neural fault injection,
- real-time anomaly detection,
- autonomous healing,
- performance recovery,
- multiclass robustness evaluation.

The healing system restored heavily degraded models to near-baseline performance levels, validating the feasibility of practical self-healing neural architectures.

These results confirm that autonomous neural recovery mechanisms can significantly improve reliability and robustness in fault-prone AI systems.

ADVANTAGES AND LIMITATIONS

Advantages

- Autonomous fault detection
- Reduced dependence on full retraining
- Improved reliability
- Modular framework design
- Supports multiple fault types
- Real-time adaptability

Limitations

- Current implementation uses a relatively simple neural architecture.
- Threshold-based detection may require tuning.
- Severe corruption may still require retraining.
- Computational overhead increases during continuous monitoring.

APPLICATIONS

The proposed framework can be applied in:

- Autonomous vehicles
- Medical diagnosis systems
- Industrial automation
- Cybersecurity systems
- Robotics
- Edge AI devices
- Fault-tolerant embedded systems

Self-healing neural networks are especially valuable in safety-critical environments where reliability is essential.

FUTURE SCOPE

Future improvements may include:

- Integration with larger datasets such as CIFAR-10 and ImageNet
- Transformer-based self-healing models
- Reinforcement learning-based healing strategies
- Autoencoder-driven anomaly detection
- Distributed self-healing systems
- Hardware-aware fault tolerance
- Real-time adaptive recovery on edge devices

The framework can also be extended to support multimodal and large-scale neural architectures.

CONCLUSION

This project presented a Self-Healing Neural Network framework capable of autonomously detecting faults and recovering degraded model performance.

A baseline neural network was developed and evaluated under multiple fault scenarios using a custom fault injection engine. The framework successfully simulated realistic neural network degradation including neuron failure, weight corruption, and layer degradation.

A monitoring and detection system was implemented to identify abnormal runtime behavior using statistical analysis of activations and prediction confidence. Automated healing strategies such as weight restoration and localized fine-tuning were applied to recover model performance.

Experimental results demonstrated that the proposed system effectively restores a significant portion of lost accuracy while maintaining operational continuity. The project highlights the feasibility of integrating self-healing capabilities into neural networks to improve reliability, robustness, and adaptability.

The work contributes toward the development of autonomous and trustworthy AI systems suitable for deployment in real-world environments.

CODE

NEURAL NETWORK

```
import os
import copy
import time
import threading
import numpy as np
import tensorflow as tf
import sklearn
from flask import Flask, jsonify, request
from flask_cors import CORS
from sklearn.metrics import (
    f1_score,    roc_auc_score,    roc_curve,    classification_report,
)
app = Flask(__name__)
CORS(app)
# GLOBAL STATE
state = {
    "status":          "idle",    "train_log":       [],
    "baseline_acc":    None,    "baseline_loss":    None,
    "epochs_done":     0,    "total_epochs":    5,
    "error":           None,    "last_faulty_acc":  None,    "last_faulty_loss": None,
    "last_healed_acc":  None,    "last_healed_loss": None,
}
model             = None
base_weights      = None
faulty_model_global = None   # Stores the most recent injected model
```

```
detector          = None    # FaultDetector instance (built after training)
healer            = None    # Healer instance (built after training)
x_test            = None
y_test            = None
loss_fn = tf.keras.losses.SparseCategoricalCrossentropy(from_logits=True)
# MODEL DEFINITION
def build_model():
    m = tf.keras.Sequential([
        tf.keras.layers.Input(shape=(28, 28)),        tf.keras.layers.Flatten(),
        tf.keras.layers.Dense(128, activation="relu"),        tf.keras.layers.Dropout(0.2),
        tf.keras.layers.Dense(10),    ])
    m.compile(optimizer="adam", loss=loss_fn, metrics=["accuracy"])
    return m
# FAULT INJECTOR
class FaultInjector:
    def __init__(self, seed=42):
        np.random.seed(seed)
    def inject(self, src_model, fault_type, severity=0.1, target_layers=None):
        corrupted = build_model()
        corrupted.set_weights(src_model.get_weights())
        layers  = corrupted.layers
        targets = target_layers if target_layers is not None else range(len(layers))
        affected = []
        for i in targets:
            if i >= len(layers):
                continue
            layer   = layers[i]
```

```
            weights = layer.get_weights()
            if not weights:
                continue
            if   fault_type == "weight_noise":       weights =
self._inject_weight_noise(weights, severity)
            elif fault_type == "weight_zero":        weights =
self._inject_weight_zero(weights, severity)
            elif fault_type == "neuron_failure":     weights =
self._inject_neuron_failure(weights, severity)
            elif fault_type == "layer_degradation":  weights =
self._inject_layer_degradation(weights, severity)
            elif fault_type == "random_swap":        weights =
self._inject_random_swap(weights)
            else: continue
            layer.set_weights(weights)
            affected.append(i)
        return corrupted, affected
    def _inject_weight_noise(self, weights, severity):
        return [w + np.random.normal(0, severity, size=w.shape) for w in weights]
    def _inject_weight_zero(self, weights, severity):
        return [w * (np.random.rand(*w.shape) > severity) for w in weights]
    def _inject_neuron_failure(self, weights, severity):
        new_weights = []
        for w in weights:
            w = w.copy()
            if w.ndim == 2:
                n      = w.shape[1]
                failed = np.random.choice(n, int(severity * n), replace=False)
                w[:, failed] = 0
```

```
            elif w.ndim == 1:
                n      = len(w)
                failed = np.random.choice(n, int(severity * n), replace=False)
                w[failed] = 0
            new_weights.append(w)
        return new_weights
    def _inject_layer_degradation(self, weights, severity):
        return [w * (1 - severity) for w in weights]
    def _inject_random_swap(self, weights):
        new_weights = []
        for w in weights:
            flat = w.flatten().copy()
            np.random.shuffle(flat)
            new_weights.append(flat.reshape(w.shape))
        return new_weights
injector = FaultInjector()
# FAULT DETECTOR
class FaultDetector:
    def __init__(self, baseline_model, baseline_data):
        self.baseline_stats = self._compute_stats(baseline_model, baseline_data)
    def _compute_stats(self, mdl, data):
        stats = []
        for layer in mdl.layers:
            try:
                if hasattr(layer, "output") and layer.output is not None:
                    sub = tf.keras.Model(inputs=mdl.inputs, outputs=layer.output)
                    act = sub(data, training=False).numpy()
```

```
                stats.append((layer.name, float(np.mean(act))))
            else:
                stats.append((layer.name, None))
        except Exception:
            stats.append((layer.name, None))
    return stats
def detect(self, check_model, data, threshold=0.2):
    current_stats = self._compute_stats(check_model, data)
    layer_diffs   = []
    is_faulty     = False
    min_len = min(len(current_stats), len(self.baseline_stats))
    for i in range(min_len):
        b_name, b_val = self.baseline_stats[i]
        c_name, c_val = current_stats[i]
        if b_val is None or c_val is None:
            continue
        diff    = abs(c_val - b_val)
        flagged = diff > threshold
        if flagged:
            is_faulty = True
        layer_diffs.append({ "layer_idx":     i,   "layer_name":    b_name,
"baseline_mean": round(b_val, 6),       "current_mean":  round(c_val, 6),
"diff":          round(diff,  6),     "flagged":       flagged,
        })
    return is_faulty, layer_diffs
# HEALER
class Healer:
    def __init__(self, original_weights):
```

```
        self.original_weights = [w.copy() for w in original_weights]
    def rollback(self, target_model, strength=0.5):
        new_weights = []
        for w, orig in zip(target_model.get_weights(), self.original_weights):
            repaired = (1.0 - strength) * w + strength * orig
            new_weights.append(repaired)
        target_model.set_weights(new_weights)
        return target_model
    def repair_layer(self, target_model, layer_idx):
        weight_offset      = 0
        layer_weight_map   = {}
        for i, layer in enumerate(target_model.layers):
            lw = layer.get_weights()
            if lw:
                layer_weight_map[i] = list(range(weight_offset, weight_offset + len(lw)))
                weight_offset += len(lw)
            else:
                layer_weight_map[i] = []
        new_weights = list(target_model.get_weights())
        for idx in layer_weight_map.get(layer_idx, []):
            new_weights[idx] = self.original_weights[idx].copy()
        target_model.set_weights(new_weights)
        return target_model
# TRAINING  (runs in background thread)
class EpochCallback(tf.keras.callbacks.Callback):
    def on_epoch_end(self, epoch, logs=None):
        state["epochs_done"] = epoch + 1
```

```
        state["train_log"].append({
            "epoch":    epoch + 1,
            "accuracy": round(float(logs.get("accuracy", 0)) * 100, 2),
            "loss":     round(float(logs.get("loss", 0)), 4),
        })
def _train_thread(epochs):
    global model, base_weights, x_test, y_test, detector, healer
    try:
        state["status"]      = "training"
        state["train_log"]   = []
        state["epochs_done"] = 0
        state["error"]       = None
        (x_train, y_train), (xt, yt) = tf.keras.datasets.mnist.load_data()
        x_train = x_train / 255.0
        x_test  = xt     / 255.0
        y_test  = yt
        m = build_model()
        m.fit(x_train, y_train, epochs=epochs, callbacks=[EpochCallback()], verbose=0)
        loss, acc = m.evaluate(x_test, y_test, verbose=0)
        model        = m
        base_weights = m.get_weights()
        detector = FaultDetector(m, x_test[:300])
        healer   = Healer(base_weights)
        state["baseline_acc"]  = round(float(acc)  * 100, 4)
        state["baseline_loss"] = round(float(loss),        4)
        state["status"]        = "ready"
    except Exception as e:
```

```
        state["status"] = "error"
        state["error"]  = str(e)
# ROUTES
@app.route("/api/status")
def api_status():
    return jsonify({
        "status":           state["status"],    "epochs_done":      state["epochs_done"],
        "total_epochs":     state["total_epochs"],        "train_log":        state["train_log"],
        "baseline_acc":     state["baseline_acc"],        "baseline_loss":
state["baseline_loss"],
        "last_faulty_acc":  state["last_faulty_acc"],        "last_faulty_loss":
state["last_faulty_loss"],
        "last_healed_acc":  state["last_healed_acc"],        "last_healed_loss":
state["last_healed_loss"],        "error":            state["error"],
    })
@app.route("/api/train", methods=["POST"])
def api_train():
    if state["status"] == "training":
        return jsonify({"error": "Already training"}), 409
    body   = request.get_json(silent=True) or {}
    epochs = int(body.get("epochs", 5))
    epochs = max(1, min(20, epochs))
    state["total_epochs"] = epochs
    t = threading.Thread(target=_train_thread, args=(epochs,), daemon=True)
    t.start()
    return jsonify({"message": f"Training started ({epochs} epochs)"})
@app.route("/api/inject", methods=["POST"])
def api_inject():
```

```
    global faulty_model_global
    if model is None:
        return jsonify({"error": "Model not trained yet"}), 400
    body = request.get_json()
    if not body:
        return jsonify({"error": "No JSON body"}), 400
    fault_type    = body.get("fault_type",    "neuron_failure")
    severity      = float(body.get("severity", 0.3))
    target_layers = body.get("target_layers", [1, 3])
    valid =
{"weight_noise","weight_zero","neuron_failure","layer_degradation","random_swap"}
    if fault_type not in valid:
        return jsonify({"error": f"Unknown fault type: {fault_type}"}), 400
    severity = max(0.0, min(1.0, severity))
    state["status"] = "injecting"
    before_weights = [w.copy() for w in model.get_weights()]
    faulty, affected = injector.inject(model, fault_type, severity, target_layers)
    faulty.compile(optimizer="adam", loss=loss_fn, metrics=["accuracy"])
    loss, acc = faulty.evaluate(x_test, y_test, verbose=0, batch_size=512)
    # Store for subsequent detect / heal calls
    faulty_model_global = faulty
    state["last_faulty_acc"]  = round(float(acc)  * 100, 4)
    state["last_faulty_loss"] = round(float(loss),        4)
    state["last_healed_acc"]  = None   # reset on new injection
    state["last_healed_loss"] = None
    after_weights = faulty.get_weights()
    weight_deltas = []
    for i, (bw, aw) in enumerate(zip(before_weights, after_weights)):
```

```
        diff = np.abs(aw - bw)
        if diff.sum() > 0:
            weight_deltas.append({
                "layer":       i,
                "mean_delta":  round(float(diff.mean()), 6),
                "max_delta":   round(float(diff.max()),  6),
                "pct_changed": round(float((diff > 0).mean() * 100), 2),
            })
    state["status"] = "ready"
    return jsonify({
        "fault_type":      fault_type,"severity": severity, "target_layers":   target_layers,
"affected_layers": affected,
        "faulty_acc":      round(float(acc)  * 100, 4), "faulty_loss":     round(float(loss),
4),  "baseline_acc":    state["baseline_acc"], "baseline_loss":   state["baseline_loss"],
        "acc_drop":        round(state["baseline_acc"] - float(acc) * 100, 4),
        "loss_increase":   round(float(loss) - state["baseline_loss"],     4),
        "weight_deltas":   weight_deltas,
    })
@app.route("/api/detect", methods=["POST"])
def api_detect():
    if model is None or detector is None:
        return jsonify({"error": "Model not trained yet"}), 400
    body        = request.get_json(silent=True) or {}
    threshold   = float(body.get("threshold",   0.2))
    sample_size = int(body.get("sample_size",   300))
    threshold   = max(0.01, min(1.0, threshold))
    sample_size = max(50,   min(1000, sample_size))
    check_model = faulty_model_global if faulty_model_global is not None else model
```

```
    sample      = x_test[:sample_size]
    state["status"] = "detecting"
    is_faulty, layer_diffs = detector.detect(check_model, sample, threshold)
    state["status"] = "ready"
    return jsonify({
        "is_faulty":    is_faulty,        "threshold":    threshold,
        "sample_size":  sample_size,   "layer_diffs":  layer_diffs,
        "faulty_acc":   state["last_faulty_acc"],   "baseline_acc": state["baseline_acc"],
    })
@app.route("/api/heal", methods=["POST"])
def api_heal():
    global faulty_model_global
    if model is None or healer is None:
        return jsonify({"error": "Model not trained yet"}), 400
    if faulty_model_global is None:
        return jsonify({"error": "No faulty model to heal — inject a fault first"}), 400
    body       = request.get_json(silent=True) or {}
    strength   = float(body.get("strength",    0.5))
    mode       = body.get("mode",              "rollback")   # "rollback" | "layer"
    layer_idx  = body.get("layer_idx",         None)
    strength = max(0.0, min(1.0, strength))
    state["status"] = "healing"
    # Snapshot faulty accuracy before healing (in case it was already modified)
    pre_heal_loss, pre_heal_acc = faulty_model_global.evaluate(
        x_test, y_test, verbose=0, batch_size=512
    )
    if mode == "layer" and layer_idx is not None:
```

```
        healer.repair_layer(faulty_model_global, int(layer_idx))
    else:
        healer.rollback(faulty_model_global, strength)
    faulty_model_global.compile(optimizer="adam", loss=loss_fn,
metrics=["accuracy"])
    loss, acc = faulty_model_global.evaluate(x_test, y_test, verbose=0, batch_size=512)
    healed_acc  = round(float(acc)  * 100, 4)
    healed_loss = round(float(loss),       4)
    state["last_healed_acc"]  = healed_acc
    state["last_healed_loss"] = healed_loss
    state["status"]           = "ready"
    faulty_acc_before = round(float(pre_heal_acc) * 100, 4)
    acc_recovery      = round(healed_acc - faulty_acc_before, 4)
    return jsonify({
        "mode":          mode,       "strength":      strength,
        "healed_acc":    healed_acc,       "healed_loss":    healed_loss,
        "baseline_acc":   state["baseline_acc"],       "baseline_loss":
state["baseline_loss"],       "faulty_acc":     faulty_acc_before,
        "acc_recovery":   acc_recovery,       "acc_remaining_gap":
round(state["baseline_acc"] - healed_acc, 4),
    })
@app.route("/api/reset", methods=["POST"])
def api_reset():
    global model, faulty_model_global
    if model is None or base_weights is None:
        return jsonify({"error": "No baseline weights available"}), 400
    model.set_weights(base_weights)
    faulty_model_global          = None
```

```
    state["last_faulty_acc"]      = None
    state["last_faulty_loss"]     = None
    state["last_healed_acc"]      = None
    state["last_healed_loss"]     = None
    return jsonify({"message": "Model reset to baseline weights"})
@app.route("/api/sample_predictions", methods=["GET"])
def api_sample_predictions():
    if model is None:
        return jsonify({"error": "Model not trained"}), 400
    check = faulty_model_global if faulty_model_global is not None else model
    prob_model = tf.keras.Sequential([check, tf.keras.layers.Softmax()])
    preds  = prob_model(x_test[:10]).numpy().tolist()
    labels = y_test[:10].tolist()
    return jsonify({
        "predictions": [
            {
                "true_label": int(labels[i]),   "pred_label": int(np.argmax(preds[i])),
                "confidence": round(float(max(preds[i])) * 100, 2),     "correct":
int(labels[i]) == int(np.argmax(preds[i])),   "probs":      [round(p, 4) for p in preds[i]],
            }
            for i in range(10)
        ]
    })
@app.route("/api/scores", methods=["POST"])
def api_scores():
    global faulty_model_global
    if model is None:
        return jsonify({"error": "Model not trained yet"}), 400
```

```
    body        = request.get_json(silent=True) or {}
    sample_size = int(body.get("sample_size", 2000))
    model_choice = body.get("model", "current")   # "baseline" | "faulty" | "current"
    sample_size = max(200, min(10000, sample_size))
    # Choose which model to score
    if model_choice == "baseline":
        score_model = model
    elif model_choice == "faulty" and faulty_model_global is not None:
        score_model = faulty_model_global
    else:
        score_model = faulty_model_global if faulty_model_global is not None else
model
    x_sample = x_test[:sample_size]
    y_true   = y_test[:sample_size]
    # Softmax probabilities
    prob_model = tf.keras.Sequential([score_model, tf.keras.layers.Softmax()])
    probs = prob_model(x_sample, training=False).numpy()   # (N, 10)
    y_pred = np.argmax(probs, axis=1)
    # F1 scores
    f1_macro  = float(f1_score(y_true, y_pred, average="macro",  zero_division=0))
    f1_micro  = float(f1_score(y_true, y_pred, average="micro",  zero_division=0))
    f1_weighted = float(f1_score(y_true, y_pred, average="weighted", zero_division=0))
    f1_per_class = f1_score(y_true, y_pred, average=None, zero_division=0).tolist()
    # ROC-AUC
    # One-hot encode true labels for multi-class AUC
    n_classes = 10
    y_true_oh = np.eye(n_classes)[y_true]
    try:
```

```
        auc_macro = float(roc_auc_score(y_true_oh, probs, average="macro",
multi_class="ovr"))
        auc_weighted = float(roc_auc_score(y_true_oh, probs, average="weighted",
multi_class="ovr"))
    except Exception:
        auc_macro    = None
        auc_weighted = None
    # Per-class AUC and ROC curves
    per_class = []
    for c in range(n_classes):
        try:
            auc_c = float(roc_auc_score(y_true_oh[:, c], probs[:, c]))
            fpr, tpr, _ = roc_curve(y_true_oh[:, c], probs[:, c])
            # Downsample curve to ≤ 50 points for JSON size
            step = max(1, len(fpr) // 50)
            per_class.append({
                "class":   c,
                "f1":      round(f1_per_class[c], 4),   "auc":     round(auc_c, 4),
                "fpr":     [round(float(v), 4) for v in fpr[::step]],   "tpr":     [round(float(v), 4)
for v in tpr[::step]],         "support": int(np.sum(y_true == c)),           })
        except Exception:
            per_class.append({
                "class": c, "f1": round(f1_per_class[c], 4),
                "auc": None, "fpr": [], "tpr": [],
                "support": int(np.sum(y_true == c)),
            })
    return jsonify({
        "model_choice":  model_choice,      "sample_size":   sample_size,
```

```
        "f1_macro":      round(f1_macro,    4),   "f1_micro":      round(f1_micro,    4),

        "f1_weighted":   round(f1_weighted, 4),      "auc_macro":     round(auc_macro,   4)
if auc_macro    is not None else None,      "auc_weighted":  round(auc_weighted,4) if
auc_weighted is not None else None,     "per_class":     per_class,      "baseline_acc":
state["baseline_acc"],

    })

@app.route("/")

def index():

    return "<p>FaultSim API running. Open <b>fault_injection_gui.html</b> in your
browser.</p>"

if __name__ == "__main__":

    print("=" * 55)

    print("  FaultSim Backend  —  http://127.0.0.1:5000")

    print("  Open fault_injection_gui.html in your browser")

    print("=" * 55)

    app.run(host="127.0.0.1", port=5000, debug=False, threaded=True)
```

HTML GUI

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>FaultSim — Neural Fault Injection Workbench</title>

<style>

@import
url('https://fonts.googleapis.com/css2?family=JetBrains+Mono:wght@300;400;500;60
0;700&family=Space+Grotesk:wght@300;400;500;600&display=swap');

:root {

  --bg:       #080c10;
```

```
  --bg2:      #0e1318;
  --bg3:      #141b22;
  --bg4:      #1a2230;
  --border:   #1f2d3d;
  --border2:  #2a3d52;
  --text:     #cdd9e5;
  --text2:    #768a9e;
  --text3:    #3a4f63;
  --cyan:     #39d0d8;
  --cyan2:    #1a9aa0;
  --green:    #3dce7a;
  --green2:   #1e6e42;
  --red:      #f05a5a;
  --red2:     #7a1f1f;
  --amber:    #e8a838;
  --amber2:   #7a4a10;
  --purple:   #a98cf4;
  --purple2:  #4a2b8c;
  --blue:     #4d9ef7;
  --mono:     'JetBrains Mono', monospace;
  --sans:     'Space Grotesk', sans-serif;
}

*{box-sizing:border-box;margin:0;padding:0;}

html{scroll-behavior:smooth;}

body{background:var(--bg);color:var(--text);font-family:var(--sans);font-size:13px;line
-height:1.55;min-height:100vh;overflow-x:hidden;}

/* GRID BACKGROUND */

body::before{
```

```
  content:'';position:fixed;inset:0;pointer-events:none;z-index:0;
  background-image:
    linear-gradient(rgba(57,208,216,.03) 1px,transparent 1px),
    linear-gradient(90deg,rgba(57,208,216,.03) 1px,transparent 1px);
  background-size:40px 40px;
}
.shell{max-width:1400px;margin:0 auto;padding:20px
24px;display:flex;flex-direction:column;gap:16px;position:relative;z-index:1;}

/* ── HEADER ── */
header{
  display:flex;align-items:center;justify-content:space-between;
  padding:14px 20px;
  background:var(--bg2);border:1px solid var(--border);border-radius:10px;
  position:relative;overflow:hidden;
}
header::before{
  content:'';position:absolute;top:0;left:0;right:0;height:2px;
  background:linear-gradient(90deg,transparent,var(--cyan),var(--purple),transparent);
}
.logo{display:flex;align-items:center;gap:12px;}
.logo-icon{
  width:38px;height:38px;border-radius:8px;
  background:linear-gradient(135deg,var(--cyan2),var(--purple2));
  display:flex;align-items:center;justify-content:center;
  box-shadow:0 0 16px rgba(57,208,216,.25);
}
.logo-text{font-family:var(--mono);font-size:16px;font-weight:700;letter-spacing:-.5px;
```

```
color:var(--text);}
.logo-sub{font-size:11px;color:var(--text2);font-family:var(--mono);margin-top:2px;}
.header-right{display:flex;align-items:center;gap:12px;}
.tech-tag{
  font-family:var(--mono);font-size:10px;color:var(--text3);
  background:var(--bg3);border:1px solid var(--border);border-radius:4px;
  padding:3px 8px;letter-spacing:.5px;
}
.status-pill{
  display:flex;align-items:center;gap:7px;
  background:var(--bg3);border:1px solid var(--border);border-radius:20px;
  padding:5px 14px;font-family:var(--mono);font-size:11px;color:var(--text2);
}
.dot{width:7px;height:7px;border-radius:50%;background:var(--green);flex-shrink:0;}
.dot.amber{background:var(--amber);}
.dot.red{background:var(--red);}
.dot.pulse{animation:pulse 1.8s ease-in-out infinite;}
@keyframes
pulse{0%,100%{opacity:1;transform:scale(1);}50%{opacity:.4;transform:scale(.8);}}
/* ── BANNER ── */
.api-banner{
  background:rgba(232,168,56,.08);border:1px solid rgba(232,168,56,.3);
  border-radius:8px;padding:10px 16px;font-family:var(--mono);font-size:12px;
  color:var(--amber);display:none;
}
.api-banner.show{display:block;}
/* ── TRAIN BAR ── */
.train-bar{
```

```
  background:var(--bg2);border:1px solid var(--border);border-radius:10px;
  padding:14px 20px;display:flex;align-items:center;gap:14px;flex-wrap:wrap;
}
.section-label{font-family:var(--mono);font-size:10px;color:var(--text3);text-transform:uppercase;letter-spacing:.8px;flex-shrink:0;}
.epoch-input{
  background:var(--bg3);border:1px solid var(--border2);border-radius:5px;
  color:var(--text);font-family:var(--mono);font-size:13px;padding:5px 10px;
  width:70px;outline:none;transition:border-color .15s;
}
.epoch-input:focus{border-color:var(--cyan);}
.progress-wrap{flex:1;min-width:200px;}
.progress-bar{height:3px;background:var(--border);border-radius:2px;overflow:hidden;}
.progress-fill{height:100%;width:0%;background:linear-gradient(90deg,var(--cyan2),var(--cyan));border-radius:2px;transition:width .4s;}
.progress-label{font-family:var(--mono);font-size:10px;color:var(--text2);margin-top:5px;}

/* ── BUTTONS ── */
.btn{
  padding:7px 16px;border-radius:6px;font-family:var(--mono);font-size:11px;
  cursor:pointer;transition:all .15s;display:inline-flex;align-items:center;gap:7px;
  border:1px solid var(--border2);background:var(--bg3);color:var(--text2);white-space:nowrap;
}
.btn:hover{border-color:var(--border2);color:var(--text);}
.btn:disabled{opacity:.35;cursor:not-allowed;}
.btn-train{border-color:var(--cyan2);color:var(--cyan);}
```

```
.btn-train:hover:not(:disabled){background:rgba(57,208,216,.08);border-color:var(--cya
n);}
.btn-inject{
  width:100%;padding:11px;justify-content:center;font-size:13px;font-weight:600;
  background:linear-gradient(135deg,var(--cyan2),rgba(57,208,216,.5));
  border:1px solid var(--cyan);color:#fff;border-radius:7px;margin-top:16px;
}
.btn-inject:hover:not(:disabled){background:linear-gradient(135deg,var(--cyan),rgba(57
,208,216,.7));}
.btn-inject:active:not(:disabled){transform:scale(.98);}
.btn-detect{
  width:100%;padding:10px;justify-content:center;font-size:12px;font-weight:600;
  background:rgba(169,140,244,.15);border:1px solid var(--purple);color:var(--purple);
  border-radius:7px;margin-top:10px;
}
.btn-detect:hover:not(:disabled){background:rgba(169,140,244,.25);}
.btn-heal{
  width:100%;padding:10px;justify-content:center;font-size:12px;font-weight:600;
  background:rgba(61,206,122,.15);border:1px solid var(--green);color:var(--green);
  border-radius:7px;margin-top:10px;
}
.btn-heal:hover:not(:disabled){background:rgba(61,206,122,.25);}
.btn-reset{border-color:transparent;color:var(--text3);}
.btn-reset:hover:not(:disabled){border-color:var(--red);color:var(--red);}
.btn-sm{padding:4px 10px;font-size:10px;}
/* ── ARCH BAR ── */
.arch-bar{
  background:var(--bg2);border:1px solid var(--border);border-radius:10px;
```

```
  padding:12px 20px;display:flex;align-items:center;gap:0;overflow-x:auto;
}
.arch-layers{display:flex;align-items:center;flex:1;gap:0;}
.arch-node{display:flex;flex-direction:column;align-items:center;flex-shrink:0;}
.arch-box{
  padding:6px 13px;border-radius:6px;border:1px solid var(--border);
  background:var(--bg3);font-family:var(--mono);font-size:11px;color:var(--text2);
  white-space:nowrap;transition:all .2s;cursor:default;
}
.arch-box.active{border-color:var(--cyan2);background:rgba(57,208,216,.07);color:var(--cyan);}
.arch-box.fault{border-color:var(--red);background:rgba(240,90,90,.1);color:var(--red);}
.arch-box.heal{border-color:var(--green);background:rgba(61,206,122,.1);color:var(--green);}
.arch-sub{font-size:10px;color:var(--text3);font-family:var(--mono);margin-top:4px;text-align:center;}
.arch-arrow{width:26px;flex-shrink:0;display:flex;align-items:center;justify-content:center;color:var(--text3);padding-bottom:16px;}
/* ── LAYOUT ── */
.main{display:grid;grid-template-columns:320px 1fr;gap:16px;align-items:start;}
/* ── PANEL ── */
.panel{background:var(--bg2);border:1px solid var(--border);border-radius:10px;overflow:hidden;margin-bottom:14px;}
.panel:last-child{margin-bottom:0;}
.panel-head{
  padding:10px 16px;border-bottom:1px solid var(--border);
  font-family:var(--mono);font-size:10px;letter-spacing:.8px;text-transform:uppercase;
  color:var(--text2);display:flex;align-items:center;gap:8px;
}
```

```
.ph-bar{width:3px;height:12px;border-radius:2px;flex-shrink:0;}
.panel-body{padding:16px;}
/* ── FAULT TYPES ── */
.fault-grid{display:grid;grid-template-columns:1fr 1fr;gap:6px;}
.fault-btn{
  padding:8px;border-radius:6px;border:1px solid var(--border);
  background:var(--bg3);color:var(--text2);font-family:var(--mono);font-size:10px;
  cursor:pointer;transition:all .15s;text-align:center;line-height:1.3;
}
.fault-btn:hover{border-color:var(--border2);color:var(--text);}
.fault-btn.active{border-color:var(--cyan);background:rgba(57,208,216,.08);color:var(--cyan);}
/* ── FIELDS ── */
.field{margin-bottom:16px;}
.field:last-child{margin-bottom:0;}
.field-label{
  font-family:var(--mono);font-size:10px;color:var(--text2);
  text-transform:uppercase;letter-spacing:.6px;margin-bottom:8px;
  display:flex;justify-content:space-between;align-items:center;
}
.field-val{font-weight:600;color:var(--cyan);font-size:13px;text-transform:none;letter-spacing:0;}
input[type=range]{
  -webkit-appearance:none;appearance:none;width:100%;height:3px;
  background:var(--border);border-radius:2px;outline:none;cursor:pointer;
}
input[type=range]::-webkit-slider-thumb{
  -webkit-appearance:none;width:15px;height:15px;border-radius:50%;
```

```
  background:var(--cyan);cursor:pointer;border:2px solid var(--bg);
  box-shadow:0 0 6px rgba(57,208,216,.5);
}
.slider-ticks{display:flex;justify-content:space-between;margin-top:5px;font-family:var(--mono);font-size:9px;color:var(--text3);}

/* ── LAYER CHECKS ── */
.layer-checks{display:flex;flex-direction:column;gap:5px;}
.layer-check{
  display:flex;align-items:center;gap:9px;
  padding:7px 10px;border-radius:6px;border:1px solid var(--border);
  background:var(--bg3);cursor:pointer;transition:all .12s;user-select:none;
}
.layer-check:hover{border-color:var(--border2);}
.layer-check.checked{border-color:rgba(57,208,216,.4);background:rgba(57,208,216,.05);}
.layer-check input{display:none;}
.chk-box{
  width:13px;height:13px;border-radius:3px;border:1px solid var(--border2);
  background:var(--bg2);display:flex;align-items:center;justify-content:center;
  flex-shrink:0;transition:all .1s;
}
.layer-check.checked .chk-box{background:var(--cyan);border-color:var(--cyan);}
.chk-tick{display:none;fill:none;}
.layer-check.checked .chk-tick{display:block;}
.chk-label{font-family:var(--mono);font-size:10px;color:var(--text2);flex:1;}
.layer-check.checked .chk-label{color:var(--text);}
.chk-badge{
```

```
  font-size:9px;padding:1px 6px;border-radius:3px;
  background:var(--bg);color:var(--text3);border:1px solid var(--border);
  font-family:var(--mono);
}
.layer-check.checked .chk-badge{color:var(--cyan);border-color:rgba(57,208,216,.3);}
/* ── HEAL CONTROLS ── */
.heal-mode-btns{display:flex;gap:6px;margin-bottom:12px;}
.heal-mode-btn{
  flex:1;padding:7px;border-radius:5px;border:1px solid var(--border);
  background:var(--bg3);color:var(--text2);font-family:var(--mono);font-size:10px;
  cursor:pointer;transition:all .12s;text-align:center;
}
.heal-mode-btn.active{border-color:var(--green);background:rgba(61,206,122,.08);colo
r:var(--green);}
.layer-idx-input{
  background:var(--bg3);border:1px solid var(--border2);border-radius:5px;
  color:var(--text);font-family:var(--mono);font-size:12px;
  padding:5px 10px;width:100%;outline:none;transition:border-color .15s;
}
.layer-idx-input:focus{border-color:var(--green);}
/* ── METRICS ROW ── */
.metrics-row{display:grid;grid-template-columns:repeat(4,1fr);gap:12px;margin-botto
m:14px;}
.mcard{
  background:var(--bg2);border:1px solid var(--border);border-radius:10px;
  padding:14px 16px;position:relative;overflow:hidden;
}
.mcard::after{content:'';position:absolute;top:0;left:0;right:0;height:2px;}
```

```
.mcard.base::after{background:var(--cyan);}

.mcard.fault-card::after{background:var(--red);}

.mcard.healed-card::after{background:var(--green);}

.mcard.delta-card::after{background:var(--amber);}

.mlabel{font-family:var(--mono);font-size:10px;color:var(--text2);text-transform:upper
case;letter-spacing:.5px;margin-bottom:6px;}

.mval{font-family:var(--mono);font-size:22px;font-weight:700;color:var(--text);line-he
ight:1;}

.mval.green{color:var(--green);}

.mval.red{color:var(--red);}

.mval.amber{color:var(--amber);}

.mval.cyan{color:var(--cyan);}

.mval.dim{color:var(--text3);letter-spacing:2px;font-size:18px;}

.msub{font-family:var(--mono);font-size:10px;color:var(--text2);margin-top:5px;}

/* ── 8-METRIC ROW ── */

.metrics-row-8{display:grid;grid-template-columns:repeat(4,1fr);gap:10px;margin-bott
om:14px;}

/* ── CHART CARD ── */

.chart-card{

  background:var(--bg2);border:1px solid var(--border);border-radius:10px;

  overflow:hidden;margin-bottom:14px;

}

.chart-card:last-child{margin-bottom:0;}

.chart-head{

  display:flex;align-items:center;justify-content:space-between;

  padding:10px 16px;border-bottom:1px solid var(--border);

}

.chart-title{font-family:var(--mono);font-size:10px;color:var(--text2);text-transform:up
percase;letter-spacing:.6px;}
```

```
.legend{display:flex;gap:12px;align-items:center;}

.leg-item{display:flex;align-items:center;gap:5px;font-size:10px;color:var(--text2);font
-family:var(--mono);}

.leg-dot{width:8px;height:8px;border-radius:2px;}

.chart-body{padding:14px;}

/* ── TABS ── */

.tabs{display:flex;gap:0;border-bottom:1px solid var(--border);padding:0
16px;background:var(--bg3);}

.tab{

  padding:9px 16px;font-family:var(--mono);font-size:10px;letter-spacing:.5px;

  text-transform:uppercase;color:var(--text2);cursor:pointer;

  border-bottom:2px solid transparent;margin-bottom:-1px;transition:all .15s;

}

.tab:hover{color:var(--text);}

.tab.active{color:var(--cyan);border-bottom-color:var(--cyan);}

.tab-content{display:none;padding:14px;}

.tab-content.active{display:block;}

/* ── DETECT TABLE ── */

.detect-table{width:100%;border-collapse:collapse;font-family:var(--mono);font-size:1
1px;}

.detect-table th{

  text-align:left;padding:6px 10px;border-bottom:1px solid var(--border);

color:var(--text2);font-size:9px;text-transform:uppercase;letter-spacing:.5px;font-weigh
t:500;

}

.detect-table td{padding:7px 10px;border-bottom:1px solid
rgba(31,45,61,.6);color:var(--text);}

.detect-table tr:last-child td{border-bottom:none;}

.detect-table tr.flagged td{background:rgba(240,90,90,.05);}
```

```
.flag-badge{
  display:inline-block;padding:1px 7px;border-radius:3px;font-size:9px;
  font-family:var(--mono);font-weight:600;
}
.flag-badge.yes{background:rgba(240,90,90,.2);color:var(--red);border:1px solid rgba(240,90,90,.3);}
.flag-badge.no{background:rgba(61,206,122,.1);color:var(--green);border:1px solid rgba(61,206,122,.2);}
.diff-bar-wrap{display:flex;align-items:center;gap:8px;}
.diff-bar{height:6px;background:var(--red);border-radius:2px;min-width:2px;transition:width .3s;}
/* ── WEIGHT DELTA TABLE ── */
.delta-table{width:100%;border-collapse:collapse;font-family:var(--mono);font-size:11px;}
.delta-table th{text-align:left;padding:6px 10px;border-bottom:1px solid var(--border);color:var(--text2);font-size:9px;text-transform:uppercase;letter-spacing:.5px;font-weight:500;}
.delta-table td{padding:7px 10px;border-bottom:1px solid rgba(31,45,61,.6);color:var(--text);}
.delta-table tr:last-child td{border-bottom:none;}
.pct-bar{display:inline-block;height:7px;background:rgba(240,90,90,.5);border-radius:2px;vertical-align:middle;margin-left:6px;}
/* ── PREDICTIONS ── */
.pred-grid{display:grid;grid-template-columns:repeat(5,1fr);gap:8px;}
.pred-card{
  background:var(--bg3);border:1px solid var(--border);border-radius:7px;
  padding:10px 8px;text-align:center;
}
.pred-card.correct{border-color:rgba(61,206,122,.35);}
.pred-card.wrong{border-color:rgba(240,90,90,.35);}
```

```
.pred-digit{font-family:var(--mono);font-size:28px;font-weight:700;line-height:1;color:
var(--text);margin-bottom:3px;}

.pred-label{font-family:var(--mono);font-size:9px;color:var(--text2);}

.pred-conf{font-family:var(--mono);font-size:10px;margin-top:3px;}

.pred-conf.correct{color:var(--green);}

.pred-conf.wrong{color:var(--red);}

.prob-bars{display:flex;flex-direction:column;gap:2px;margin-top:6px;}

.prob-bar-row{display:flex;align-items:center;gap:4px;font-family:var(--mono);font-siz
e:8px;color:var(--text3);}

.prob-bar-fill{height:4px;border-radius:1px;background:var(--cyan2);transition:width
.3s;}

/* ── DETECT STATUS ── */

.detect-status{

  display:flex;align-items:center;gap:12px;padding:12px 16px;

  border-radius:8px;margin-bottom:12px;border:1px solid;

}

.detect-status.fault{background:rgba(240,90,90,.07);border-color:rgba(240,90,90,.3);}

.detect-status.ok{background:rgba(61,206,122,.07);border-color:rgba(61,206,122,.3);}

.detect-status.pending{background:rgba(57,208,216,.07);border-color:rgba(57,208,216,.
2);}

.ds-icon{width:28px;height:28px;border-radius:50%;display:flex;align-items:center;jus
tify-content:center;flex-shrink:0;font-size:14px;}

.detect-status.fault .ds-icon{background:rgba(240,90,90,.2);}

.detect-status.ok .ds-icon{background:rgba(61,206,122,.2);}

.ds-text{font-family:var(--mono);font-size:12px;}

.ds-main{font-weight:600;}

.detect-status.fault .ds-main{color:var(--red);}

.detect-status.ok .ds-main{color:var(--green);}

.ds-sub{font-size:10px;color:var(--text2);margin-top:2px;}
```

```
/* ── HEAL RESULTS ── */
.heal-result-bar{
  padding:12px 16px;border-radius:8px;border:1px solid;margin-bottom:12px;
}
.heal-result-bar.success{background:rgba(61,206,122,.07);border-color:rgba(61,206,122,.3);}
.heal-grid{display:grid;grid-template-columns:repeat(3,1fr);gap:10px;}
.heal-stat{text-align:center;padding:10px;background:var(--bg3);border-radius:7px;border:1px solid var(--border);}
.heal-stat-label{font-family:var(--mono);font-size:9px;color:var(--text2);text-transform:uppercase;letter-spacing:.5px;margin-bottom:4px;}
.heal-stat-val{font-family:var(--mono);font-size:17px;font-weight:700;}
.heal-stat-val.green{color:var(--green);}
.heal-stat-val.amber{color:var(--amber);}
.heal-stat-val.red{color:var(--red);}
/* ── FAULT DESC ── */
.fault-desc-box{
  padding:12px;background:var(--bg3);border-radius:7px;border:1px solid var(--border);
  font-size:11px;color:var(--text2);line-height:1.6;
}
.fault-desc-title{color:var(--cyan);font-family:var(--mono);font-size:10px;font-weight:600;margin-bottom:5px;text-transform:uppercase;letter-spacing:.5px;}
/* ── LOG ── */
.log-card{background:var(--bg);border:1px solid var(--border);border-radius:10px;overflow:hidden;}
.log-header{
  display:flex;align-items:center;justify-content:space-between;
  padding:9px 14px;border-bottom:1px solid var(--border);background:var(--bg2);
}
```

```
.log-title{font-family:var(--mono);font-size:10px;color:var(--text2);text-transform:uppe
rcase;letter-spacing:.6px;}
.log-body{
  padding:10px 14px;font-family:var(--mono);font-size:11px;line-height:1.9;
  height:150px;overflow-y:auto;color:var(--text2);
}
.log-entry{display:flex;gap:10px;}
.log-time{color:var(--text3);flex-shrink:0;font-size:10px;}
.log-msg.info{color:var(--cyan);}
.log-msg.ok{color:var(--green);}
.log-msg.warn{color:var(--amber);}
.log-msg.err{color:var(--red);}
.log-msg.dim{color:var(--text3);}
.log-msg.purple{color:var(--purple);}
/* ── TOAST ── */
.toast{
  position:fixed;bottom:24px;right:24px;
  background:var(--bg2);border:1px solid var(--border);border-radius:9px;
  padding:12px 18px;font-family:var(--mono);font-size:12px;color:var(--text);
  z-index:200;transform:translateY(80px);opacity:0;transition:all .3s;max-width:360px;
  box-shadow:0 8px 32px rgba(0,0,0,.4);
}
.toast.show{transform:translateY(0);opacity:1;}
.toast.err{border-color:var(--red);color:var(--red);}
.toast.ok{border-color:var(--green);color:var(--green);}
.toast.purple{border-color:var(--purple);color:var(--purple);}
/* ── SCROLLBAR ── */
::-webkit-scrollbar{width:4px;height:4px;}
```

```
::-webkit-scrollbar-track{background:transparent;}
::-webkit-scrollbar-thumb{background:var(--border2);border-radius:2px;}
.spin{display:inline-block;animation:spin .6s linear infinite;}
@keyframes spin{to{transform:rotate(360deg);}}
/* ── SCORES TAB ── */
.score-cards{display:grid;grid-template-columns:repeat(4,1fr);gap:10px;margin-bottom:14px;}
.score-card{
  background:var(--bg3);border:1px solid var(--border);border-radius:8px;
  padding:12px 14px;position:relative;overflow:hidden;
}
.score-card::after{content:'';position:absolute;top:0;left:0;right:0;height:2px;}
.score-card.f1::after{background:var(--blue);}
.score-card.auc::after{background:var(--purple);}
.sc-label{font-family:var(--mono);font-size:9px;color:var(--text2);text-transform:uppercase;letter-spacing:.6px;margin-bottom:5px;}
.sc-val{font-family:var(--mono);font-size:20px;font-weight:700;line-height:1;}
.sc-val.blue{color:var(--blue);}
.sc-val.purple{color:var(--purple);}
.sc-val.green{color:var(--green);}
.sc-val.amber{color:var(--amber);}
.sc-val.red{color:var(--red);}
.sc-val.dim{color:var(--text3);font-size:16px;letter-spacing:2px;}
.sc-sub{font-family:var(--mono);font-size:9px;color:var(--text2);margin-top:4px;}
.score-model-btns{display:flex;gap:6px;margin-bottom:14px;flex-wrap:wrap;}
.score-model-btn{
  padding:6px 14px;border-radius:5px;border:1px solid var(--border);
  background:var(--bg3);color:var(--text2);font-family:var(--mono);font-size:10px;
```

```
  cursor:pointer;transition:all .12s;
}
.score-model-btn.active{border-color:var(--blue);background:rgba(77,158,247,.1);color
:var(--blue);}
.score-model-btn:hover:not(.active){border-color:var(--border2);color:var(--text);}
.per-class-table{width:100%;border-collapse:collapse;font-family:var(--mono);font-size
:11px;}
.per-class-table th{text-align:left;padding:5px 8px;border-bottom:1px solid
var(--border);color:var(--text2);font-size:9px;text-transform:uppercase;letter-spacing:.5
px;font-weight:500;}
.per-class-table td{padding:6px 8px;border-bottom:1px solid
rgba(31,45,61,.5);color:var(--text);}
.per-class-table tr:last-child td{border-bottom:none;}
.mini-bar{display:inline-block;height:6px;border-radius:2px;vertical-align:middle;marg
in-left:5px;}


.section-divider{
  display:flex;align-items:center;gap:12px;
  font-family:var(--mono);font-size:9px;color:var(--text3);
  text-transform:uppercase;letter-spacing:.8px;margin:10px 0;
}
.section-divider::before,.section-divider::after{content:'';flex:1;height:1px;background:v
ar(--border);}
/* ── EMPTY STATES ── */
.empty-state{
  text-align:center;padding:30px 20px;
  font-family:var(--mono);font-size:11px;color:var(--text3);
}
.empty-icon{font-size:24px;margin-bottom:8px;}
</style>
```

```
</head>
<body>
<div class="shell">
<!-- HEADER -->
<header>
  <div class="logo">
    <div class="logo-icon">
      <svg width="20" height="20" viewBox="0 0 20 20" fill="none">
        <circle cx="10" cy="10" r="3" fill="rgba(255,255,255,.9)"/>
        <circle cx="3" cy="3" r="2" fill="rgba(255,255,255,.5)"/>
        <circle cx="17" cy="3" r="2" fill="rgba(255,255,255,.5)"/>
        <circle cx="3" cy="17" r="2" fill="rgba(255,255,255,.5)"/>
        <circle cx="17" cy="17" r="2" fill="rgba(255,255,255,.5)"/>
        <line x1="5" y1="3" x2="8" y2="8.5" stroke="rgba(255,255,255,.4)"
stroke-width="1"/>
        <line x1="15" y1="3" x2="12" y2="8.5" stroke="rgba(255,255,255,.4)"
stroke-width="1"/>
        <line x1="5" y1="17" x2="8" y2="11.5" stroke="rgba(255,255,255,.4)"
stroke-width="1"/>
        <line x1="15" y1="17" x2="12" y2="11.5" stroke="rgba(255,255,255,.4)"
stroke-width="1"/>
      </svg>
    </div>
    <div>
      <div class="logo-text">FaultSim</div>
      <div class="logo-sub">Neural Fault Injection · Detection · Healing
Workbench</div>
    </div>
  </div>
```

```
  <div class="header-right">
    <span class="tech-tag">MNIST · TensorFlow · Flask</span>
    <div class="status-pill"><div class="dot pulse" id="status-dot"></div><span
id="status-text">connecting…</span></div>
  </div>
</header>
<!-- API BANNER -->
<div class="api-banner" id="api-banner">
  ⚠ Cannot reach backend at http://127.0.0.1:5000 — ensure <code>python
app.py</code> is running
</div>
<!-- TRAIN BAR -->
<div class="train-bar">
  <span class="section-label">Training</span>
  <input class="epoch-input" type="number" id="epoch-input" min="1" max="20"
value="5">
  <span class="section-label">epochs</span>
  <button class="btn btn-train" id="train-btn">▶ Train Model</button>
  <div class="progress-wrap">
    <div class="progress-bar"><div class="progress-fill"
id="progress-fill"></div></div>
    <div class="progress-label" id="progress-label">awaiting training…</div>
  </div>
  <button class="btn" id="pred-btn" disabled>◈ Sample Predictions</button>
  <button class="btn" id="scores-btn-top" disabled
style="border-color:var(--blue);color:var(--blue);">⬡ F1 &amp;amp; AUC</button>
  <button class="btn btn-reset" id="reset-btn" disabled>↺ Reset Weights</button>
</div>
<!-- ARCH BAR -->
```

```
<div class="arch-bar">
  <span class="section-label" style="margin-right:14px;">Architecture</span>
  <div class="arch-layers">
    <div class="arch-node">
      <div class="arch-box">Input</div>
      <div class="arch-sub">28×28</div>
    </div>
    <div class="arch-arrow">→</div>
    <div class="arch-node">
      <div class="arch-box" id="arch-0">Flatten</div>
      <div class="arch-sub">784</div>
    </div>
    <div class="arch-arrow">→</div>
    <div class="arch-node">
      <div class="arch-box" id="arch-1">Dense</div>
      <div class="arch-sub">128 · ReLU</div>
    </div>
    <div class="arch-arrow">→</div>
    <div class="arch-node">
      <div class="arch-box" id="arch-2">Dropout</div>
      <div class="arch-sub">p=0.2</div>
    </div>
    <div class="arch-arrow">→</div>
    <div class="arch-node">
      <div class="arch-box" id="arch-3">Dense</div>
      <div class="arch-sub">10 · Logits</div>
    </div>
```

```
    <div class="arch-arrow">→</div>
    <div class="arch-node">
      <div class="arch-box">Softmax</div>
      <div class="arch-sub">Output</div>
    </div>
  </div>
</div>

<!-- MAIN LAYOUT -->
<div class="main">
  <!-- LEFT COLUMN: CONTROLS -->
  <div>
    <!-- FAULT INJECTION PANEL -->
    <div class="panel">
      <div class="panel-head">
        <div class="ph-bar" style="background:var(--red);"></div>
        Fault Injection
      </div>
      <div class="panel-body">
        <div class="field">
          <div class="field-label">Fault Type</div>
          <div class="fault-grid">
            <div class="fault-btn active" data-fault="neuron_failure">neuron_failure</div>
            <div class="fault-btn" data-fault="weight_noise">weight_noise</div>
            <div class="fault-btn" data-fault="weight_zero">weight_zero</div>
            <div class="fault-btn" data-fault="layer_degradation">layer_degradation</div>
            <div class="fault-btn" data-fault="random_swap" style="grid-column:span
2;">random_swap</div>
```

```
      </div>

    </div>

    <div class="field">

      <div class="field-label">Severity <span class="field-val"
id="sev-val">0.30</span></div>

      <input type="range" id="severity" min="0.05" max="1.0" step="0.05"
value="0.30">

      <div
class="slider-ticks"><span>0.05</span><span>0.25</span><span>0.50</span><span>
0.75</span><span>1.0</span></div>

    </div>

    <div class="field">

      <div class="field-label">Target Layers</div>

      <div class="layer-checks">

        <label class="layer-check" id="lc-0">

          <input type="checkbox" data-layer="0">

          <div class="chk-box"><svg class="chk-tick" width="9" height="9"
viewBox="0 0 9 9"><polyline points="1.5,4.5 3.5,6.5 7.5,2" stroke="white"
stroke-width="1.5" stroke-linecap="round" stroke-linejoin="round"/></svg></div>

          <div class="chk-label">L0 — Flatten</div>

          <div class="chk-badge">L0</div>

        </label>

        <label class="layer-check checked" id="lc-1">

          <input type="checkbox" checked data-layer="1">

          <div class="chk-box"><svg class="chk-tick" width="9" height="9"
viewBox="0 0 9 9"><polyline points="1.5,4.5 3.5,6.5 7.5,2" stroke="white"
stroke-width="1.5" stroke-linecap="round" stroke-linejoin="round"/></svg></div>

          <div class="chk-label">L1 — Dense 128</div>

          <div class="chk-badge">L1</div>

        </label>
```

```
          <label class="layer-check" id="lc-2">
            <input type="checkbox" data-layer="2">
            <div class="chk-box"><svg class="chk-tick" width="9" height="9" viewBox="0 0 9 9"><polyline points="1.5,4.5 3.5,6.5 7.5,2" stroke="white" stroke-width="1.5" stroke-linecap="round" stroke-linejoin="round"/></svg></div>
            <div class="chk-label">L2 — Dropout</div>
            <div class="chk-badge">L2</div>
          </label>
          <label class="layer-check checked" id="lc-3">
            <input type="checkbox" checked data-layer="3">
            <div class="chk-box"><svg class="chk-tick" width="9" height="9" viewBox="0 0 9 9"><polyline points="1.5,4.5 3.5,6.5 7.5,2" stroke="white" stroke-width="1.5" stroke-linecap="round" stroke-linejoin="round"/></svg></div>
            <div class="chk-label">L3 — Dense 10</div>
            <div class="chk-badge">L3</div>
          </label>
        </div>
      </div>
      <div class="fault-desc-box" id="fault-desc-box">
        <div class="fault-desc-title" id="fd-title">neuron_failure</div>
        <div id="fd-text">Zeroes out a random subset of neurons (columns) in weight matrices. Simulates hardware neuron stuck-at-zero faults.</div>
      </div>
      <button class="btn btn-inject" id="inject-btn" disabled>⚡ Inject Fault</button>
    </div>
  </div>
  <!-- DETECTION PANEL -->
  <div class="panel">
    <div class="panel-head">
```

```
      <div class="ph-bar" style="background:var(--purple);"></div>
      Fault Detection
    </div>
    <div class="panel-body">
      <div class="field">
        <div class="field-label">Threshold <span class="field-val"
id="thresh-val">0.20</span></div>
        <input type="range" id="threshold" min="0.01" max="1.0" step="0.01"
value="0.20">
        <div
class="slider-ticks"><span>0.01</span><span>0.25</span><span>0.50</span><span>
0.75</span><span>1.0</span></div>
      </div>
      <div class="field">
        <div class="field-label">Sample Size <span class="field-val"
id="sample-val">300</span></div>
        <input type="range" id="sample-size" min="50" max="1000" step="50"
value="300">
        <div
class="slider-ticks"><span>50</span><span>300</span><span>600</span><span>10
00</span></div>
      </div>
      <button class="btn btn-detect" id="detect-btn" disabled>🔍 Run
Detection</button>
    </div>
  </div>
  <!-- HEALING PANEL -->
  <div class="panel">
    <div class="panel-head">
      <div class="ph-bar" style="background:var(--green);"></div>
      Model Healing
```

```
    </div>
    <div class="panel-body">
      <div class="field-label" style="margin-bottom:8px;">Heal Mode</div>
      <div class="heal-mode-btns">
        <button class="heal-mode-btn active" data-mode="rollback"
id="hm-rollback">Rollback Blend</button>
        <button class="heal-mode-btn" data-mode="layer" id="hm-layer">Layer
Repair</button>
      </div>
      <div id="heal-rollback-controls">
        <div class="field">
          <div class="field-label">Blend Strength <span class="field-val"
id="strength-val">0.50</span></div>
          <input type="range" id="heal-strength" min="0.0" max="1.0" step="0.05"
value="0.50">
          <div
class="slider-ticks"><span>0.0</span><span>0.25</span><span>0.50</span><span>0
.75</span><span>1.0</span></div>
        </div>
        <div
style="font-size:10px;color:var(--text2);font-family:var(--mono);line-height:1.6;">
          <span style="color:var(--green);">1.0</span> = full restore to baseline
&amp;nbsp;·&amp;nbsp; <span style="color:var(--amber);">0.0</span> = no change
        </div>
      </div>
      <div id="heal-layer-controls" style="display:none;">
        <div class="field">
          <div class="field-label">Layer Index to Repair</div>
          <input class="layer-idx-input" type="number" id="layer-idx" min="0" max="3"
value="1" placeholder="0–3">
        </div>
```

```
        <div style="font-size:10px;color:var(--text2);font-family:var(--mono);">Fully restores one layer to its original baseline weights.</div>
      </div>
      <button class="btn btn-heal" id="heal-btn" disabled>✦ Heal Model</button>
    </div>
  </div>
</div><!-- /left -->
<!-- RIGHT COLUMN: RESULTS -->
<div>
  <!-- METRICS: ALL 8 -->
  <div class="metrics-row-8">
    <div class="mcard base">
      <div class="mlabel">Baseline Acc</div>
      <div class="mval dim" id="base-acc">—</div>
      <div class="msub" id="base-acc-sub">train first</div>
    </div>
    <div class="mcard base">
      <div class="mlabel">Baseline Loss</div>
      <div class="mval dim" id="base-loss">—</div>
      <div class="msub" id="base-loss-sub">train first</div>
    </div>
    <div class="mcard fault-card">
      <div class="mlabel">Faulty Acc</div>
      <div class="mval dim" id="faulty-acc">—</div>
      <div class="msub" id="faulty-acc-sub">inject fault</div>
    </div>
    <div class="mcard fault-card">
      <div class="mlabel">Faulty Loss</div>
```

```
    <div class="mval dim" id="faulty-loss">—</div>
    <div class="msub" id="faulty-loss-sub">inject fault</div>
  </div>
  <div class="mcard delta-card">
    <div class="mlabel">Acc Drop</div>
    <div class="mval dim" id="acc-drop">—</div>
    <div class="msub" id="acc-drop-sub">pp from baseline</div>
  </div>
  <div class="mcard delta-card">
    <div class="mlabel">Loss Δ</div>
    <div class="mval dim" id="loss-delta">—</div>
    <div class="msub" id="loss-delta-sub">from baseline</div>
  </div>
  <div class="mcard healed-card">
    <div class="mlabel">Healed Acc</div>
    <div class="mval dim" id="healed-acc">—</div>
    <div class="msub" id="healed-acc-sub">heal model</div>
  </div>
  <div class="mcard healed-card">
    <div class="mlabel">Recovery</div>
    <div class="mval dim" id="acc-recovery">—</div>
    <div class="msub" id="acc-recovery-sub">pp recovered</div>
  </div>
</div>
<!-- TABBED RESULTS AREA -->
<div class="chart-card">
  <div class="tabs" id="results-tabs">
```

```
      <div class="tab active" data-tab="history">Run History</div>
      <div class="tab" data-tab="detect">Detection</div>
      <div class="tab" data-tab="heal">Heal Results</div>
      <div class="tab" data-tab="deltas">Weight Deltas</div>
      <div class="tab" data-tab="scores">F1 &amp;amp; ROC-AUC</div>
      <div class="tab" data-tab="predictions">Predictions</div>
    </div>
    <!-- TAB: RUN HISTORY -->
    <div class="tab-content active" id="tab-history">
      <div style="display:grid;grid-template-columns:1fr 1fr;gap:12px;">
        <div>
          <div
style="font-family:var(--mono);font-size:10px;color:var(--text2);text-transform:upperc
ase;letter-spacing:.5px;margin-bottom:8px;display:flex;justify-content:space-between;"
>
            <span>Accuracy — run history</span>
            <span style="display:flex;gap:10px;">
              <span style="display:flex;align-items:center;gap:4px;"><span
class="leg-dot" style="background:#39d0d8;"></span>Baseline</span>
              <span style="display:flex;align-items:center;gap:4px;"><span
class="leg-dot" style="background:#f05a5a;"></span>Faulty</span>
              <span style="display:flex;align-items:center;gap:4px;"><span
class="leg-dot" style="background:#3dce7a;"></span>Healed</span>
            </span>
          </div>
          <div style="position:relative;height:210px;"><canvas
id="accChart"></canvas></div>
        </div>
        <div>
          <div
```

```
style="font-family:var(--mono);font-size:10px;color:var(--text2);text-transform:upperc
ase;letter-spacing:.5px;margin-bottom:8px;display:flex;justify-content:space-between;"
>

          <span>Loss — run history</span>

          <span style="display:flex;gap:10px;">

            <span style="display:flex;align-items:center;gap:4px;"><span
class="leg-dot" style="background:#39d0d8;"></span>Baseline</span>

            <span style="display:flex;align-items:center;gap:4px;"><span
class="leg-dot" style="background:#e8a838;"></span>Faulty</span>

            <span style="display:flex;align-items:center;gap:4px;"><span
class="leg-dot" style="background:#3dce7a;"></span>Healed</span>

          </span>

        </div>

        <div style="position:relative;height:210px;"><canvas
id="lossChart"></canvas></div>

      </div>

    </div>

    <div id="history-empty" class="empty-state">

      <div class="empty-icon">📊</div>

      <div>Inject a fault to see run history</div>

    </div>

    <div id="history-table-wrap" style="display:none;margin-top:14px;">

      <table class="detect-table" id="history-table">

        <thead><tr><th>#</th><th>Fault
Type</th><th>Sev</th><th>Layers</th><th>Base Acc</th><th>Faulty
Acc</th><th>Healed Acc</th><th>Recovery</th></tr></thead>

        <tbody id="history-tbody"></tbody>

      </table>

    </div>

  </div>
```

```
    <!-- TAB: DETECTION -->
    <div class="tab-content" id="tab-detect">
      <div id="detect-empty" class="empty-state">
        <div class="empty-icon">🔍</div>
        <div>Run detection after injecting a fault</div>
      </div>
      <div id="detect-results" style="display:none;">
        <div class="detect-status pending" id="detect-status-bar">
          <div class="ds-icon">?</div>
          <div class="ds-text">
            <div class="ds-main" id="ds-main-text">No detection run yet</div>
            <div class="ds-sub" id="ds-sub-text">Run detection to analyse layer activations</div>
          </div>
        </div>
        <table class="detect-table" id="detect-table">
          <thead><tr><th>#</th><th>Layer</th><th>Baseline Mean</th><th>Current Mean</th><th>Diff</th><th>Threshold</th><th>Status</th></tr></thead>
          <tbody id="detect-tbody"></tbody>
        </table>
      </div>
    </div>
    <!-- TAB: HEAL RESULTS -->
    <div class="tab-content" id="tab-heal">
      <div id="heal-empty" class="empty-state">
        <div class="empty-icon">✦</div>
        <div>Heal the model after fault injection</div>
      </div>
```

```
      <div id="heal-results" style="display:none;">
        <div class="heal-result-bar success" id="heal-result-bar">
          <div
style="font-family:var(--mono);font-size:11px;color:var(--green);font-weight:600;marg
in-bottom:10px;" id="heal-mode-label">Rollback Blend — strength 0.50</div>
          <div class="heal-grid">
            <div class="heal-stat">
              <div class="heal-stat-label">Pre-Heal Acc</div>
              <div class="heal-stat-val amber" id="h-pre-acc">—</div>
            </div>
            <div class="heal-stat">
              <div class="heal-stat-label">Post-Heal Acc</div>
              <div class="heal-stat-val green" id="h-post-acc">—</div>
            </div>
            <div class="heal-stat">
              <div class="heal-stat-label">Recovery</div>
              <div class="heal-stat-val green" id="h-recovery">—</div>
            </div>
            <div class="heal-stat">
              <div class="heal-stat-label">Baseline Acc</div>
              <div class="heal-stat-val" id="h-base-acc"
style="color:var(--cyan);">—</div>
            </div>
            <div class="heal-stat">
              <div class="heal-stat-label">Healed Loss</div>
              <div class="heal-stat-val" id="h-post-loss"
style="color:var(--text);">—</div>
            </div>
            <div class="heal-stat">
```

```
            <div class="heal-stat-label">Remaining Gap</div>
            <div class="heal-stat-val amber" id="h-gap">—</div>
          </div>
        </div>
      </div>
      <div style="position:relative;height:180px;margin-top:12px;">
        <canvas id="healChart"></canvas>
      </div>
    </div>
  </div>
  <!-- TAB: WEIGHT DELTAS -->
  <div class="tab-content" id="tab-deltas">
    <div id="deltas-empty" class="empty-state">
      <div class="empty-icon">⚖</div>
      <div>Weight delta analysis appears after fault injection</div>
    </div>
    <div id="deltas-results" style="display:none;">
      <table class="delta-table" id="delta-table">
        <thead><tr><th>Weight Tensor</th><th>Mean |Δw|</th><th>Max |Δw|</th><th>% Changed</th></tr></thead>
        <tbody id="delta-body"></tbody>
      </table>
    </div>
  </div>
  <!-- TAB: F1 & ROC-AUC SCORES -->
  <div class="tab-content" id="tab-scores">
    <!-- Model selector -->
    <div
```

```
style="display:flex;align-items:center;justify-content:space-between;margin-bottom:12
px;flex-wrap:wrap;gap:8px;">
        <div class="score-model-btns">
          <button class="score-model-btn active" data-smodel="current">Current
Model</button>
          <button class="score-model-btn" data-smodel="baseline">Baseline</button>
          <button class="score-model-btn" data-smodel="faulty">Faulty</button>
        </div>
        <button class="btn" id="scores-btn" disabled
style="border-color:var(--blue);color:var(--blue);">⬡ Compute Scores</button>
      </div>
      <div id="scores-empty" class="empty-state">
        <div class="empty-icon">⬡</div>
        <div>Train the model then click "Compute Scores" to evaluate F1 &amp;amp;
ROC-AUC</div>
      </div>
      <div id="scores-results" style="display:none;">
        <!-- Summary metric cards -->
        <div class="score-cards">
          <div class="score-card f1">
            <div class="sc-label">F1 Macro</div>
            <div class="sc-val dim" id="sc-f1-macro">—</div>
            <div class="sc-sub" id="sc-f1-macro-sub">averaged over 10 classes</div>
          </div>
          <div class="score-card f1">
            <div class="sc-label">F1 Weighted</div>
            <div class="sc-val dim" id="sc-f1-weighted">—</div>
            <div class="sc-sub" id="sc-f1-weighted-sub">support-weighted</div>
          </div>
```

```
        <div class="score-card auc">
          <div class="sc-label">ROC-AUC Macro</div>
          <div class="sc-val dim" id="sc-auc-macro">—</div>
          <div class="sc-sub" id="sc-auc-macro-sub">OvR macro average</div>
        </div>
        <div class="score-card auc">
          <div class="sc-label">ROC-AUC Weighted</div>
          <div class="sc-val dim" id="sc-auc-weighted">—</div>
          <div class="sc-sub" id="sc-auc-weighted-sub">support-weighted OvR</div>
        </div>
      </div>
      <!-- Charts: ROC curve + F1 bar -->
      <div style="display:grid;grid-template-columns:1fr
1fr;gap:12px;margin-bottom:14px;">
        <div>
          <div
style="font-family:var(--mono);font-size:10px;color:var(--text2);text-transform:upperc
ase;letter-spacing:.5px;margin-bottom:8px;display:flex;justify-content:space-between;"
>
            <span>ROC Curves — all 10 classes</span>
            <span style="color:var(--text3);">higher = better</span>
          </div>
          <div style="position:relative;height:220px;"><canvas
id="rocChart"></canvas></div>
        </div>
        <div>
          <div
style="font-family:var(--mono);font-size:10px;color:var(--text2);text-transform:upperc
ase;letter-spacing:.5px;margin-bottom:8px;display:flex;justify-content:space-between;"
>
```

```
            <span>Per-Class F1 Score</span>
            <span style="color:var(--text3);">by digit 0–9</span>
          </div>
          <div style="position:relative;height:220px;"><canvas
id="f1BarChart"></canvas></div>
        </div>
      </div>
      <!-- Per-class breakdown table -->
      <div
style="font-family:var(--mono);font-size:10px;color:var(--text2);text-transform:upperc
ase;letter-spacing:.5px;margin-bottom:8px;">Per-Class Breakdown</div>
      <table class="per-class-table" id="per-class-table">
        <thead>
          <tr>
            <th>Class</th>
            <th>Support</th>
            <th>F1 Score</th>
            <th>F1 Bar</th>
            <th>ROC-AUC</th>
            <th>AUC Bar</th>
          </tr>
        </thead>
        <tbody id="per-class-tbody"></tbody>
      </table>
    </div>
  </div>
  <!-- TAB: PREDICTIONS -->
  <div class="tab-content" id="tab-predictions">
```

```
      <div id="pred-empty" class="empty-state">
        <div class="empty-icon">◈</div>
        <div>Click "Sample Predictions" to view model outputs on test images</div>
      </div>
      <div id="pred-results" style="display:none;">
        <div
style="display:flex;align-items:center;justify-content:space-between;margin-bottom:10
px;">
          <span style="font-family:var(--mono);font-size:10px;color:var(--text2);">First
10 test images — current model state</span>
          <span id="pred-score"
style="font-family:var(--mono);font-size:11px;color:var(--cyan);"></span>
        </div>
        <div class="pred-grid" id="pred-grid"></div>
      </div>
    </div>
  </div><!-- chart-card -->
 </div><!-- /right -->
</div><!-- /main -->
<!-- LOG -->
<div class="log-card">
 <div class="log-header">
  <span class="log-title">API Console</span>
  <button class="btn btn-sm" id="clear-log">Clear</button>
 </div>
 <div class="log-body" id="log"></div>
</div>
</div><!-- .shell -->
<div class="toast" id="toast"></div>
```

```
<script
src="https://cdnjs.cloudflare.com/ajax/libs/Chart.js/4.4.1/chart.umd.js"></script>

<script>

const API = 'http://127.0.0.1:5000';

// ——— Global State
————————————————————————————————————————————————————————————

let runHistory = [];

let accChart, lossChart, healChart;

let selectedFault = 'neuron_failure';

let healMode = 'rollback';

let lastDetectResult = null;

let lastHealResult = null;

const faultDescs = {

  neuron_failure:    'Zeroes out a random subset of neurons (columns) in weight
matrices. Simulates hardware neuron stuck-at-zero faults.',

  weight_noise:      'Adds Gaussian noise N(0, severity) to all weights. Models analog
hardware noise and quantisation errors.',

  weight_zero:       'Randomly masks weights to zero with probability = severity.
Simulates stuck-at-zero bit faults in memory cells.',

  layer_degradation: 'Scales all weights in target layers by (1 − severity). Models
gradual hardware wear or voltage degradation.',

  random_swap:       'Shuffles all weight values randomly within each tensor. Models
catastrophic bit-flip or data corruption events.'

};

// ——— Utilities
————————————————————————————————————————————————————————————
—

function nowStr(){const d=new Date();return d.toTimeString().slice(0,8);}


function log(msg, cls=''){

  const el = document.getElementById('log');
```

```
  const e = document.createElement('div');
  e.className = 'log-entry';
  e.innerHTML = `<span class="log-time">${nowStr()}</span><span class="log-msg
${cls}">${msg}</span>`;
  el.appendChild(e);
  el.scrollTop = el.scrollHeight;
}
function toast(msg, type=''){
  const t = document.getElementById('toast');
  t.textContent = msg;
  t.className = 'toast show' + (type ? ' ' + type : '');
  clearTimeout(t._timer);
  t._timer = setTimeout(() => t.classList.remove('show'), 3500);
}
function setStatus(text, dotClass=''){
  document.getElementById('status-text').textContent = text;
  const dot = document.getElementById('status-dot');
  dot.className = 'dot' + (dotClass ? ' ' + dotClass : ' pulse');
}
function fmtAcc(v){ return v != null ? v.toFixed(2) + '%' : '—'; }
function fmtLoss(v){ return v != null ? v.toFixed(4) : '—'; }

function setMval(id, text, colorClass=''){
  const el = document.getElementById(id);
  el.textContent = text;
  el.className = 'mval' + (colorClass ? ' ' + colorClass : '');
}
function switchTab(tabId){
```

```
  document.querySelectorAll('.tab').forEach(t => t.classList.toggle('active', t.dataset.tab
=== tabId));

  document.querySelectorAll('.tab-content').forEach(tc => tc.classList.toggle('active',
tc.id === 'tab-' + tabId));

}

// ——— API Fetch
————————————————————————————————————————————————————————————
——

async function apiFetch(path, opts={}){

  try {

    const r = await fetch(API + path, {headers: {'Content-Type': 'application/json'},
...opts});

    if(!r.ok){

      const errBody = await r.json().catch(() => ({}));

      throw new Error(errBody.error || `HTTP ${r.status}`);

    }

    return await r.json();

  } catch(e) {

    document.getElementById('api-banner').classList.add('show');

    throw e;

  }

}


// ——— Charts
————————————————————————————————————————————————————————————
————————

function chartDefaults(){

  return {

    responsive: true, maintainAspectRatio: false,

    plugins: {
```

```
    legend: {display: false},
    tooltip: {
      backgroundColor: '#0e1318', borderColor: '#1f2d3d', borderWidth: 1,
      titleColor: '#768a9e', bodyColor: '#cdd9e5',
      titleFont: {family: 'JetBrains Mono', size: 10},
      bodyFont: {family: 'JetBrains Mono', size: 11}
    }
  },
  scales: {
    x: {grid: {color: '#111c27'}, ticks: {color: '#768a9e', font: {family: 'JetBrains Mono', size: 9}}},
    y: {grid: {color: '#111c27'}, ticks: {color: '#768a9e', font: {family: 'JetBrains Mono', size: 9}}}
  },
  animation: {duration: 300}
 };
}
function initCharts(){
 const cd = chartDefaults();

 accChart = new Chart(document.getElementById('accChart'), {
  type: 'bar',
  data: {labels: [], datasets: [
    {label: 'Baseline', data: [], backgroundColor: 'rgba(57,208,216,.6)', borderColor: '#39d0d8', borderWidth: 1, borderRadius: 3},
    {label: 'Faulty',   data: [], backgroundColor: 'rgba(240,90,90,.6)',  borderColor: '#f05a5a', borderWidth: 1, borderRadius: 3},
    {label: 'Healed',   data: [], backgroundColor: 'rgba(61,206,122,.6)', borderColor: '#3dce7a', borderWidth: 1, borderRadius: 3},
```

```
  ]},
  options: {...cd, scales: {...cd.scales, y: {...cd.scales.y, min: 0, max: 100, ticks:
{...cd.scales.y.ticks, callback: v => v + '%'}}}}
 });
 lossChart = new Chart(document.getElementById('lossChart'), {
  type: 'bar',
  data: {labels: [], datasets: [
    {label: 'Baseline', data: [], backgroundColor: 'rgba(57,208,216,.6)',  borderColor:
'#39d0d8', borderWidth: 1, borderRadius: 3},
    {label: 'Faulty',   data: [], backgroundColor: 'rgba(232,168,56,.6)',  borderColor:
'#e8a838', borderWidth: 1, borderRadius: 3},
    {label: 'Healed',   data: [], backgroundColor: 'rgba(61,206,122,.6)',  borderColor:
'#3dce7a', borderWidth: 1, borderRadius: 3},
  ]},
  options: {...cd, scales: {...cd.scales, y: {...cd.scales.y, min: 0, ticks:
{...cd.scales.y.ticks, callback: v => v.toFixed(3)}}}}
 });
 healChart = new Chart(document.getElementById('healChart'), {
  type: 'bar',
  data: {
    labels: ['Faulty Acc', 'Healed Acc', 'Baseline Acc'],
    datasets: [{
      data: [0, 0, 0],
      backgroundColor: ['rgba(240,90,90,.65)', 'rgba(61,206,122,.65)',
'rgba(57,208,216,.65)'],
      borderColor: ['#f05a5a', '#3dce7a', '#39d0d8'],
      borderWidth: 1, borderRadius: 4
    }]
  },
  options: {...cd, scales: {...cd.scales, y: {...cd.scales.y, min: 0, max: 100, ticks:
```

```
{...cd.scales.y.ticks, callback: v => v + '%'}}}}
  });
}
function updateHistoryCharts(){
  const last = runHistory.slice(-8);
  const labels = last.map((r, i) => `#${runHistory.indexOf(r)+1} ${r.fault.slice(0,6)}`);
  accChart.data.labels = labels;
  accChart.data.datasets[0].data = last.map(r => r.base_acc);
  accChart.data.datasets[1].data = last.map(r => r.faulty_acc);
  accChart.data.datasets[2].data = last.map(r => r.healed_acc ?? null);
  accChart.update();
  lossChart.data.labels = labels;
  lossChart.data.datasets[0].data = last.map(r => r.base_loss);
  lossChart.data.datasets[1].data = last.map(r => r.faulty_loss);
  lossChart.data.datasets[2].data = last.map(r => r.healed_loss ?? null);
  lossChart.update();
}
function updateHealChart(faultyAcc, healedAcc, baselineAcc){
  healChart.data.datasets[0].data = [faultyAcc, healedAcc, baselineAcc];
  healChart.update();
}
// ─── Architecture Highlight ──────────────────────────────────────
function highlightArch(layers, mode='fault'){
  [0,1,2,3].forEach(i => {
    const el = document.getElementById('arch-' + i);
    if(el) el.className = 'arch-box';
  });
```

```
  (layers || []).forEach(i => {
    const el = document.getElementById('arch-' + i);
    if(el) el.className = 'arch-box ' + mode;
  });
}
// ——— History Table
function updateHistoryTable(){
  const empty = document.getElementById('history-empty');
  const wrap = document.getElementById('history-table-wrap');
  if(!runHistory.length){ empty.style.display = ''; wrap.style.display = 'none'; return; }
  empty.style.display = 'none'; wrap.style.display = '';
  const tbody = document.getElementById('history-tbody');
  tbody.innerHTML = runHistory.map((r, i) => `
    <tr>
      <td>${i+1}</td>
      <td style="color:var(--red)">${r.fault}</td>
      <td>${r.severity}</td>
      <td>[${r.layers.join(',')}]</td>
      <td style="color:var(--cyan)">${fmtAcc(r.base_acc)}</td>
      <td style="color:var(--red)">${fmtAcc(r.faulty_acc)}</td>
      <td style="color:${r.healed_acc != null ? 'var(--green)' :
'var(--text3)'}">${r.healed_acc != null ? fmtAcc(r.healed_acc) : '—'}</td>
      <td style="color:var(--amber)">${r.healed_acc != null ? '+' + (r.healed_acc -
r.faulty_acc).toFixed(2) + 'pp' : '—'}</td>
    </tr>`).join('');
}
// ——— Poll Training
```

```
async function pollStatus(){
  const fill = document.getElementById('progress-fill');
  const lbl = document.getElementById('progress-label');
  return new Promise(resolve => {
    const iv = setInterval(async () => {
      try {
        const s = await apiFetch('/api/status');
        const pct = s.total_epochs > 0 ? (s.epochs_done / s.total_epochs * 100) : 0;
        fill.style.width = pct + '%';
        if(s.train_log && s.train_log.length){
          const last = s.train_log[s.train_log.length - 1];
          lbl.textContent = `Epoch ${last.epoch}/${s.total_epochs} — acc:
${last.accuracy}%  loss: ${last.loss}`;
        }
        if(s.status === 'ready'){ clearInterval(iv); fill.style.width = '100%'; resolve(s); }
        else if(s.status === 'error'){ clearInterval(iv); resolve(s); }
      } catch(e){ clearInterval(iv); resolve({status: 'error', error: String(e)}); }
    }, 800);
  });
}
// ——— TRAIN
————————————————————————————————————————————————————————————

async function doTrain(){
  const epochs = parseInt(document.getElementById('epoch-input').value) || 5;

['train-btn','inject-btn','detect-btn','heal-btn','reset-btn','pred-btn','scores-btn','scores-btn-t
op'].forEach(id => {
    const el = document.getElementById(id);
```

```
  if(el) el.disabled = true;
 });
 setStatus('training…', 'amber');
 log(`POST /api/train  epochs=${epochs}`, 'info');
 try {
  await apiFetch('/api/train', {method: 'POST', body: JSON.stringify({epochs})});
  log('Training started — polling…', 'dim');
  const s = await pollStatus();
  if(s.status === 'error'){
   log('Training error: ' + s.error, 'err');
   toast('Training failed: ' + s.error, 'err');
   setStatus('error', 'red'); return;
  }
  // Update baseline metrics
  setMval('base-acc',  s.baseline_acc + '%', 'cyan');
  setMval('base-loss', s.baseline_loss, '');
  document.getElementById('base-acc-sub').textContent = epochs + ' epochs · Adam';
  document.getElementById('base-loss-sub').textContent = 'SparseCatCrossentropy';
  document.getElementById('api-banner').classList.remove('show');
  setStatus('ready', 'green');
  ['inject-btn','reset-btn','pred-btn','scores-btn','scores-btn-top'].forEach(id =>
document.getElementById(id).disabled = false);
  highlightArch([0,1,2,3], 'active');
  log(`Training complete — baseline acc=${s.baseline_acc}%
loss=${s.baseline_loss}`, 'ok');
  toast('Model ready — acc ' + s.baseline_acc + '%', 'ok');
 } catch(e){
  log('Error: ' + e.message, 'err');
```

```
    setStatus('error', 'red');
  } finally {
    document.getElementById('train-btn').disabled = false;
  }
}
// ——— INJECT ———————————————————————————————————————————————————————

async function doInject(){
  const severity = parseFloat(document.getElementById('severity').value);
  const target_layers = [...document.querySelectorAll('.layer-check
input:checked')].map(e => +e.dataset.layer);
  if(!target_layers.length){ toast('Select at least one target layer', 'err'); return; }
  const btn = document.getElementById('inject-btn');
  btn.disabled = true;
  btn.innerHTML = '<span class="spin">↻</span> Injecting…';
  document.getElementById('detect-btn').disabled = true;
  document.getElementById('heal-btn').disabled = true;
  setStatus('injecting…', 'amber');
  highlightArch(target_layers, 'fault');
  log(`POST /api/inject  fault=${selectedFault}  severity=${severity}
layers=[${target_layers}]`, 'info');
  try {
    const res = await apiFetch('/api/inject', {
      method: 'POST',
      body: JSON.stringify({fault_type: selectedFault, severity, target_layers})
    });
    // Faulty metrics
    const accColorClass = res.faulty_acc < 60 ? 'red' : res.faulty_acc < 85 ? 'amber' : '';
```

```
    setMval('faulty-acc',  res.faulty_acc + '%', accColorClass);
    setMval('faulty-loss', res.faulty_loss, '');
    document.getElementById('faulty-acc-sub').textContent  = `↓
${res.acc_drop.toFixed(2)}pp from baseline`;
    document.getElementById('faulty-loss-sub').textContent = `↑
${res.loss_increase.toFixed(4)} from baseline`;


    // Delta metrics
    setMval('acc-drop',   '−' + res.acc_drop.toFixed(2) + 'pp', 'red');
    setMval('loss-delta', '+' + res.loss_increase.toFixed(4), 'amber');
    document.getElementById('acc-drop-sub').textContent   = 'accuracy degradation';
    document.getElementById('loss-delta-sub').textContent = 'loss increase';
    // Reset heal metrics on new injection
    setMval('healed-acc',   '—', 'dim');
    setMval('acc-recovery', '—', 'dim');
    document.getElementById('healed-acc-sub').textContent   = 'heal model';
    document.getElementById('acc-recovery-sub').textContent = 'pp recovered';
    // Weight deltas tab
    if(res.weight_deltas && res.weight_deltas.length){
      document.getElementById('deltas-empty').style.display = 'none';
      document.getElementById('deltas-results').style.display = '';
      const tbody = document.getElementById('delta-body');
      tbody.innerHTML = res.weight_deltas.map(d => `
        <tr>
          <td style="color:var(--text2);">tensor L${d.layer}</td>
          <td>${d.mean_delta.toFixed(6)}</td>
          <td>${d.max_delta.toFixed(6)}</td>
          <td>${d.pct_changed.toFixed(1)}%<span class="pct-bar"
```

```
style="width:${Math.min(d.pct_changed * 2, 150)}px;"></span></td>
      </tr>`).join('');
  }

  // History
  runHistory.push({
    fault: selectedFault, severity, layers: target_layers,
    base_acc: res.baseline_acc, base_loss: res.baseline_loss,
    faulty_acc: res.faulty_acc, faulty_loss: res.faulty_loss,
    healed_acc: null, healed_loss: null,
    affected: res.affected_layers
  });
  updateHistoryCharts();
  updateHistoryTable();
  // Enable detect & heal
  document.getElementById('detect-btn').disabled = false;
  document.getElementById('heal-btn').disabled = false;
  setStatus('ready — fault injected', 'amber');
  log(`  → faulty acc=${res.faulty_acc}%  loss=${res.faulty_loss}
Δacc=−${res.acc_drop.toFixed(2)}pp`, res.faulty_acc > 85 ? 'warn' : 'err');
  toast(`Fault injected — acc dropped to ${res.faulty_acc}%`, res.faulty_acc < 60 ?
'err' : '');
  // Auto-switch to history tab
  switchTab('history');
 } catch(e){
  log('Injection error: ' + e.message, 'err');
  toast('Injection failed', 'err');
  setStatus('error', 'red');
```

```
    highlightArch([0,1,2,3], 'active');
  } finally {
    btn.disabled = false;
    btn.innerHTML = '⚡ Inject Fault';
  }
}
// ——— DETECT ———————————————————————————————————————————————————————————————————————————————

async function doDetect(){
  const threshold = parseFloat(document.getElementById('threshold').value);
  const sample_size = parseInt(document.getElementById('sample-size').value);
  const btn = document.getElementById('detect-btn');
  btn.disabled = true;
  btn.innerHTML = '<span class="spin">↻</span> Detecting…';
  setStatus('detecting…', 'amber');
  log(`POST /api/detect  threshold=${threshold}  sample_size=${sample_size}`,
'purple');
  try {
    const res = await apiFetch('/api/detect', {
      method: 'POST',
      body: JSON.stringify({threshold, sample_size})
    });
    lastDetectResult = res;
    switchTab('detect');

    // Status bar
    const statusBar = document.getElementById('detect-status-bar');
```

```
    const dsMain = document.getElementById('ds-main-text');
    const dsSub = document.getElementById('ds-sub-text');
    if(res.is_faulty){
      statusBar.className = 'detect-status fault';
      statusBar.querySelector('.ds-icon').textContent = '⚠';
      dsMain.textContent = 'FAULT DETECTED';
      dsSub.textContent = `Activation anomalies found · threshold=${threshold} · ${sample_size} samples`;
    } else {
      statusBar.className = 'detect-status ok';
      statusBar.querySelector('.ds-icon').textContent = '✓';
      dsMain.textContent = 'No significant faults detected';
      dsSub.textContent = `All layers within threshold=${threshold} · ${sample_size} samples`;
    }
    // Layer diffs table
    document.getElementById('detect-empty').style.display = 'none';
    document.getElementById('detect-results').style.display = '';
    const tbody = document.getElementById('detect-tbody');
    tbody.innerHTML = res.layer_diffs.map(d => `
      <tr class="${d.flagged ? 'flagged' : ''}">
        <td>${d.layer_idx}</td>
        <td style="color:var(--text2);font-family:var(--mono);font-size:10px;">${d.layer_name}</td>
        <td>${d.baseline_mean.toFixed(5)}</td>
        <td>${d.current_mean.toFixed(5)}</td>
        <td>
          <div class="diff-bar-wrap">
```

```
        <span>${d.diff.toFixed(5)}</span>

        <div class="diff-bar" style="width:${Math.min(d.diff / threshold * 60,
90)}px;background:${d.flagged ? 'var(--red)' : 'var(--green)'};"></div>

       </div>

      </td>

      <td style="color:var(--text2);">${threshold}</td>

      <td><span class="flag-badge ${d.flagged ? 'yes' : 'no'}">${d.flagged ?
'FLAGGED' : 'OK'}</span></td>

    </tr>`).join('');

  const flagCount = res.layer_diffs.filter(d => d.flagged).length;

  setStatus('ready', 'green');

  log(`  → is_faulty=${res.is_faulty}  flagged=${flagCount}/${res.layer_diffs.length}
layers`, res.is_faulty ? 'err' : 'ok');

  toast(res.is_faulty ? `Fault detected — ${flagCount} layer(s) anomalous` :
'Detection: no faults found', res.is_faulty ? 'err' : 'ok');

 } catch(e){

  log('Detection error: ' + e.message, 'err');

  toast('Detection failed', 'err');

  setStatus('error', 'red');

 } finally {

  btn.disabled = false;

  btn.innerHTML = '🔍 Run Detection';

 }

}


// ─── HEAL
──────────────────────────────────────────────────────────────────────
──────────

async function doHeal(){

 const strength = parseFloat(document.getElementById('heal-strength').value);
```

```
  const layer_idx = healMode === 'layer' ?
parseInt(document.getElementById('layer-idx').value) : null;

  const btn = document.getElementById('heal-btn');

  btn.disabled = true;

  btn.innerHTML = '<span class="spin">↻</span> Healing…';

  setStatus('healing…', 'amber');

  const body = {mode: healMode, strength};

  if(healMode === 'layer') body.layer_idx = layer_idx;

  log(`POST /api/heal  mode=${healMode}  strength=${strength}${healMode ===
'layer' ? '  layer_idx='+layer_idx : ''}`, 'ok');

  try {

    const res = await apiFetch('/api/heal', {method: 'POST', body:
JSON.stringify(body)});

    lastHealResult = res;

    // Update heal metrics

    const healColor = res.healed_acc >= 85 ? 'green' : res.healed_acc >= 70 ? 'amber' :
'red';

    setMval('healed-acc',   res.healed_acc + '%', healColor);

    setMval('acc-recovery', '+' + res.acc_recovery.toFixed(2) + 'pp', 'green');

    document.getElementById('healed-acc-sub').textContent   = `gap to baseline:
${res.acc_remaining_gap.toFixed(2)}pp`;

    document.getElementById('acc-recovery-sub').textContent = healMode === 'layer' ?
`layer ${layer_idx} restored` : `strength=${strength}`;


    // Heal results tab

    document.getElementById('heal-empty').style.display = 'none';

    document.getElementById('heal-results').style.display = '';

    document.getElementById('heal-mode-label').textContent =

      healMode === 'rollback'

        ? `Rollback Blend — strength ${strength}`
```

```
      : `Layer Repair — layer ${layer_idx} fully restored`;
    document.getElementById('h-pre-acc').textContent  = fmtAcc(res.faulty_acc);
    document.getElementById('h-post-acc').textContent = fmtAcc(res.healed_acc);
    document.getElementById('h-recovery').textContent = '+' +
res.acc_recovery.toFixed(2) + 'pp';
    document.getElementById('h-base-acc').textContent = fmtAcc(res.baseline_acc);
    document.getElementById('h-post-loss').textContent = fmtLoss(res.healed_loss);
    document.getElementById('h-gap').textContent = res.acc_remaining_gap.toFixed(2)
+ 'pp';
    // Update heal chart
    updateHealChart(res.faulty_acc, res.healed_acc, res.baseline_acc);
    // Update history
    if(runHistory.length){
      const last = runHistory[runHistory.length - 1];
      last.healed_acc = res.healed_acc;
      last.healed_loss = res.healed_loss;
      updateHistoryCharts();
      updateHistoryTable();
    }
    // Highlight arch in heal color
    highlightArch(healMode === 'layer' ? [layer_idx] : [1, 3], 'heal');
    switchTab('heal');
    setStatus('ready — healed', 'green');
    log(`  → healed acc=${res.healed_acc}%  loss=${res.healed_loss}
recovery=+${res.acc_recovery.toFixed(2)}pp`, 'ok');
    toast(`Healed! Acc recovered to ${res.healed_acc}%
(+${res.acc_recovery.toFixed(2)}pp)`, 'ok');
  } catch(e){
    log('Heal error: ' + e.message, 'err');
```

```
    toast('Healing failed: ' + e.message, 'err');
    setStatus('error', 'red');
  } finally {
    btn.disabled = false;
    btn.innerHTML = '✦ Heal Model';
  }
}
// ——— RESET ———————————————
```

```
    // Reset detect/heal UI
    document.getElementById('detect-empty').style.display = '';
    document.getElementById('detect-results').style.display = 'none';
    document.getElementById('heal-empty').style.display = '';
    document.getElementById('heal-results').style.display = 'none';
    document.getElementById('deltas-empty').style.display = '';
    document.getElementById('deltas-results').style.display = 'none';
    setStatus('ready', 'green');
  } catch(e){
    log('Reset error: ' + e.message, 'err');
  }
}
// ——— SAMPLE PREDICTIONS ————————————————————————————————————

async function doPredict(){
  const btn = document.getElementById('pred-btn');
  btn.disabled = true;
  btn.innerHTML = '<span class="spin">↻</span> Loading…';
  log('GET /api/sample_predictions', 'info');
  try {
    const res = await apiFetch('/api/sample_predictions');
    const grid = document.getElementById('pred-grid');
    const correct = res.predictions.filter(p => p.correct).length;
    document.getElementById('pred-score').textContent = `${correct}/10 correct`;
    grid.innerHTML = res.predictions.map(p => {
      const topProbs = p.probs
        .map((v, i) => ({v, i}))
        .sort((a, b) => b.v - a.v)
```

```
      .slice(0, 3);
    return `
      <div class="pred-card ${p.correct ? 'correct' : 'wrong'}">
        <div class="pred-digit">${p.pred_label}</div>
        <div class="pred-label">true: ${p.true_label}</div>
        <div class="pred-conf ${p.correct ? 'correct' : 'wrong'}">${p.confidence.toFixed(1)}%</div>
        <div class="prob-bars">
          ${topProbs.map(t => `
            <div class="prob-bar-row">
              <span style="width:10px;">${t.i}</span>
              <div class="prob-bar-fill" style="width:${Math.round(t.v * 60)}px;background:${t.i === p.true_label ? 'var(--green)' : t.i === p.pred_label && !p.correct ? 'var(--red)' : 'var(--cyan2)'}"></div>
              <span>${(t.v * 100).toFixed(0)}%</span>
            </div>`).join('')}
        </div>
      </div>`;
  }).join('');
  document.getElementById('pred-empty').style.display = 'none';
  document.getElementById('pred-results').style.display = '';
  switchTab('predictions');
  log(`  → ${correct}/10 correct`, correct >= 8 ? 'ok' : 'warn');
 } catch(e){
  log('Predictions error: ' + e.message, 'err');
  toast('Failed to load predictions', 'err');
 } finally {
  btn.disabled = false;
```

```
    btn.innerHTML = '◈ Sample Predictions';
  }
}
// ——— CONNECTION CHECK ———————————————————————————————————

async function checkConnection(){
  try {
    const s = await apiFetch('/api/status');
    document.getElementById('api-banner').classList.remove('show');
    if(s.status === 'ready'){
      setStatus('ready', 'green');
      setMval('base-acc',  s.baseline_acc + '%', 'cyan');
      setMval('base-loss', s.baseline_loss, '');
      document.getElementById('base-acc-sub').textContent  = 'model loaded';
      document.getElementById('base-loss-sub').textContent = 'SparseCatCrossentropy';
      ['inject-btn','reset-btn','pred-btn','scores-btn','scores-btn-top'].forEach(id =>
document.getElementById(id).disabled = false);
      if(s.last_faulty_acc != null){
        setMval('faulty-acc',  s.last_faulty_acc + '%', s.last_faulty_acc < 60 ? 'red' :
'amber');
        setMval('faulty-loss', s.last_faulty_loss, '');
        document.getElementById('detect-btn').disabled = false;
        document.getElementById('heal-btn').disabled = false;
      }
      if(s.last_healed_acc != null){
        setMval('healed-acc',  s.last_healed_acc + '%', 'green');
      }
      highlightArch([0,1,2,3], 'active');
```

```
      log('Backend connected — model already trained and ready', 'ok');
    } else {
      setStatus('connected', 'green');
      log('Backend connected — click "Train Model" to begin', 'info');
    }
  } catch(e){
    setStatus('offline', 'red');
    document.getElementById('api-banner').classList.add('show');
    log('Backend not reachable — is app.py running on port 5000?', 'err');
  }
}
// ─── SCORES: F1 & ROC-AUC ──────────────────────────────────────────

let rocChart = null;
let f1BarChart = null;
let selectedScoreModel = 'current';
const CLASS_COLORS = [
  '#39d0d8','#3dce7a','#f05a5a','#e8a838','#a98cf4',
  '#4d9ef7','#f07050','#70d0a8','#d870d0','#c8c040'
];
function initScoreCharts(){
  const cd = chartDefaults();
  rocChart = new Chart(document.getElementById('rocChart'), {
    type: 'line',
    data: {labels: [], datasets: []},
    options: {
      ...cd,
      elements: {point: {radius: 0}},
```

```
    scales: {
      x: {...cd.scales.x, min: 0, max: 1, title: {display: true, text: 'FPR', color: '#768a9e', font: {family: 'JetBrains Mono', size: 9}}},
      y: {...cd.scales.y, min: 0, max: 1, title: {display: true, text: 'TPR', color: '#768a9e', font: {family: 'JetBrains Mono', size: 9}}}
    },
    plugins: {
      ...cd.plugins,
      legend: {
        display: true, position: 'right',
        labels: {color: '#768a9e', font: {family: 'JetBrains Mono', size: 9}, boxWidth: 10, padding: 8}
      }
    }
  }
});
f1BarChart = new Chart(document.getElementById('f1BarChart'), {
  type: 'bar',
  data: {
    labels: ['0','1','2','3','4','5','6','7','8','9'],
    datasets: [{
      label: 'F1 Score',
      data: new Array(10).fill(0),
      backgroundColor: CLASS_COLORS.map(c => c + 'aa'),
      borderColor: CLASS_COLORS,
      borderWidth: 1, borderRadius: 3
    }]
  },
```

```
    options: {
      ...cd,
      scales: {
        x: {...cd.scales.x},
        y: {...cd.scales.y, min: 0, max: 1, ticks: {...cd.scales.y.ticks, callback: v =>
v.toFixed(1)}}
      }
    }
  });
}
function scColor(v, type='f1'){
  if(v == null) return 'dim';
  if(type === 'auc') return v >= 0.98 ? 'green' : v >= 0.95 ? 'blue' : v >= 0.9 ? 'amber' :
'red';
  return v >= 0.95 ? 'green' : v >= 0.85 ? 'blue' : v >= 0.70 ? 'amber' : 'red';
}

function setSc(id, val, type='f1'){
  const el = document.getElementById(id);
  if(!el) return;
  if(val == null){ el.textContent = '—'; el.className = 'sc-val dim'; return; }
  el.textContent = val.toFixed(4);
  el.className = 'sc-val ' + scColor(val, type);
}
async function doScores(){
  const btn1 = document.getElementById('scores-btn');
  const btn2 = document.getElementById('scores-btn-top');
  [btn1, btn2].forEach(b => { if(b){ b.disabled = true; b.innerHTML = '<span
```

```
class="spin">↻</span> Computing…'; } });
  log(`POST /api/scores  model=${selectedScoreModel}`, 'purple');
  try {
    const res = await apiFetch('/api/scores', {
      method: 'POST',
      body: JSON.stringify({model: selectedScoreModel, sample_size: 2000})
    });
    // Show results
    document.getElementById('scores-empty').style.display = 'none';
    document.getElementById('scores-results').style.display = '';
    switchTab('scores');
    // Summary cards
    setSc('sc-f1-macro',    res.f1_macro,    'f1');
    setSc('sc-f1-weighted', res.f1_weighted, 'f1');
    setSc('sc-auc-macro',   res.auc_macro,   'auc');
    setSc('sc-auc-weighted',res.auc_weighted,'auc');
    document.getElementById('sc-f1-macro-sub').textContent    = `model:
${res.model_choice} · n=${res.sample_size}`;
    document.getElementById('sc-f1-weighted-sub').textContent  = `micro:
${res.f1_micro.toFixed(4)}`;
    document.getElementById('sc-auc-macro-sub').textContent    = 'OvR macro · 10
classes';
    document.getElementById('sc-auc-weighted-sub').textContent = 'OvR
support-weighted';
    // ROC curve chart — one dataset per class
    rocChart.data.datasets = res.per_class.map((pc, i) => ({
      label: `${pc.class} (AUC ${pc.auc != null ? pc.auc.toFixed(3) : 'N/A'})`,
      data: pc.fpr.map((x, j) => ({x, y: pc.tpr[j]})),
      borderColor: CLASS_COLORS[i],
```

```
    backgroundColor: 'transparent',
    borderWidth: 1.5,
    tension: 0.1,
    parsing: false
  }));
  // Diagonal reference line
  rocChart.data.datasets.push({
    label: 'Random',
    data: [{x:0,y:0},{x:1,y:1}],
    borderColor: 'rgba(255,255,255,.1)',
    borderDash: [4,4],
    borderWidth: 1,
    pointRadius: 0,
    parsing: false
  });
  rocChart.update();
  // F1 bar chart
  f1BarChart.data.datasets[0].data = res.per_class.map(pc => pc.f1);
  f1BarChart.update();
  // Per-class table
  const tbody = document.getElementById('per-class-tbody');
  tbody.innerHTML = res.per_class.map(pc => {
    const f1Color = pc.f1 >= 0.95 ? 'var(--green)' : pc.f1 >= 0.85 ? 'var(--blue)' : pc.f1
>= 0.70 ? 'var(--amber)' : 'var(--red)';
    const aucColor = pc.auc == null ? 'var(--text3)' : pc.auc >= 0.99 ? 'var(--green)' :
pc.auc >= 0.97 ? 'var(--blue)' : 'var(--amber)';
    return `
      <tr>
```

```
        <td><span
style="display:inline-block;width:8px;height:8px;border-radius:2px;background:${CL
ASS_COLORS[pc.class]};margin-right:7px;vertical-align:middle;"></span>Digit
${pc.class}</td>

        <td style="color:var(--text2);">${pc.support}</td>

        <td style="color:${f1Color};font-weight:600;">${pc.f1.toFixed(4)}</td>

        <td><span class="mini-bar"
style="width:${Math.round(pc.f1*80)}px;background:${f1Color};"></span></td>

        <td style="color:${aucColor};font-weight:600;">${pc.auc != null ?
pc.auc.toFixed(4) : '—'}</td>

        <td>${pc.auc != null ? `<span class="mini-bar"
style="width:${Math.round((pc.auc-0.5)*160)}px;background:${aucColor};"></span>
` : '—'}</td>

       </tr>`;

    }).join('');

    log(`  → F1 macro=${res.f1_macro.toFixed(4)}  AUC macro=${res.auc_macro !=
null ? res.auc_macro.toFixed(4) : 'N/A'}`, 'purple');

    toast(`Scores: F1=${res.f1_macro.toFixed(3)}  AUC=${res.auc_macro != null ?
res.auc_macro.toFixed(3) : 'N/A'}`, 'purple');

  } catch(e){

    log('Scores error: ' + e.message, 'err');

    toast('Score computation failed: ' + e.message, 'err');

  } finally {

    [btn1, btn2].forEach(b => {

      if(b){ b.disabled = false; b.innerHTML = b.id === 'scores-btn' ? '⬡ Compute
Scores' : '⬡ F1 &amp;amp; AUC'; }

    });

  }

}

// ——— EVENT LISTENERS
═══════════════════════════════════════════════════════════════

document.querySelectorAll('.fault-btn').forEach(btn => {
```

```
  btn.addEventListener('click', () => {
    document.querySelectorAll('.fault-btn').forEach(b => b.classList.remove('active'));
    btn.classList.add('active');
    selectedFault = btn.dataset.fault;
    document.getElementById('fd-title').textContent = selectedFault;
    document.getElementById('fd-text').textContent  = faultDescs[selectedFault];
  });
});
document.getElementById('severity').addEventListener('input', e => {
  document.getElementById('sev-val').textContent =
parseFloat(e.target.value).toFixed(2);
});
document.getElementById('threshold').addEventListener('input', e => {
  document.getElementById('thresh-val').textContent =
parseFloat(e.target.value).toFixed(2);
});
document.getElementById('sample-size').addEventListener('input', e => {
  document.getElementById('sample-val').textContent = e.target.value;
});
document.getElementById('heal-strength').addEventListener('input', e => {
  document.getElementById('strength-val').textContent =
parseFloat(e.target.value).toFixed(2);
});
document.querySelectorAll('.layer-check').forEach(lc => {
  lc.addEventListener('click', () => {
    const cb = lc.querySelector('input');
    cb.checked = !cb.checked;
    lc.classList.toggle('checked', cb.checked);
```

```
  });
});
document.querySelectorAll('.heal-mode-btn').forEach(btn => {
  btn.addEventListener('click', () => {
    document.querySelectorAll('.heal-mode-btn').forEach(b =>
b.classList.remove('active'));
    btn.classList.add('active');
    healMode = btn.dataset.mode;
    document.getElementById('heal-rollback-controls').style.display = healMode ===
'rollback' ? '' : 'none';
    document.getElementById('heal-layer-controls').style.display    = healMode ===
'layer'    ? '' : 'none';
  });
});
document.querySelectorAll('.tab').forEach(tab => {
  tab.addEventListener('click', () => switchTab(tab.dataset.tab));
});
document.getElementById('train-btn').addEventListener('click', doTrain);
document.getElementById('inject-btn').addEventListener('click', doInject);
document.getElementById('detect-btn').addEventListener('click', doDetect);
document.getElementById('heal-btn').addEventListener('click', doHeal);
document.getElementById('reset-btn').addEventListener('click', doReset);
document.getElementById('pred-btn').addEventListener('click', doPredict);
document.getElementById('scores-btn').addEventListener('click', doScores);
document.getElementById('scores-btn-top').addEventListener('click', doScores);
document.getElementById('clear-log').addEventListener('click', () =>
document.getElementById('log').innerHTML = '');
document.querySelectorAll('.score-model-btn').forEach(btn => {
  btn.addEventListener('click', () => {
```

```
    document.querySelectorAll('.score-model-btn').forEach(b =>
b.classList.remove('active'));
    btn.classList.add('active');
    selectedScoreModel = btn.dataset.smodel;
  });
});
// ——— Init
————————————————————————————————————————————————————————————

initCharts();
initScoreCharts();
log('FaultSim GUI loaded — checking backend connection…', 'dim');
checkConnection();
</script>
</body>
</html>
```